Electromagnetism — Problem Sheet 1

Worked solutions to the extension questions (Problems 1–20)

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Questions and corrections are welcome by email.

This document solves the extension questions that close each problem in the worked solutions to Sheet 1. For each problem, a green box first lists the background the extensions rely on, flagging anything that goes beyond the main problem or the earlier parts of the course. Each extension is then restated in a shaded box, followed by a step-by-step solution with the algebra written out and, where useful, a short comment on what it teaches.

Symbols follow the main solutions: ε_0 is the permittivity of free space, μ_0 the permeability of free space, c the speed of light, $Z_0 = \sqrt{\mu_0/\varepsilon_0} = 377\,\Omega$ the impedance of free space, and $k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^9\,\mathrm{m\,F^{-1}}$. The toolkit boxes at the start of each set of main solutions (binomial expansion, polar coordinates and their unit vectors, complex notation, plane waves) are used freely. Every numerical result has been checked by independent computation.

Problem 1 — Finite line of charge

What you need to know for these extensions

- The set-up of Problem 1: Coulomb's law for an element $\lambda\,\mathrm{d}x$, with $\mathbf{r}$ the vector from the element to the field point, and the substitution $x = z\tan\alpha$.
- **Sum-to-product identities** (only for the optional remark in 1.1): $\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$ and $\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$.
- The binomial expansion, including $(1+u)^{-3/2} \approx 1 - \frac{3}{2}u$.

1.1 Field at a general point

Find $\mathbf{E}$ *at a general point off the bisector; express it in terms of the angles subtended by the two ends of the segment.*

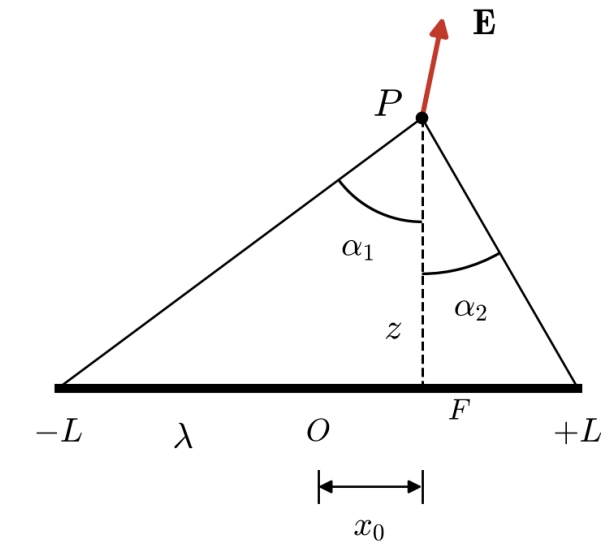


Figure 1: A general field point P, at height z above the line and axial position x_0; F is the foot of the perpendicular from P. The angles to the two ends are measured from PF, positive towards $+x$, so here $\alpha_1 < 0 < \alpha_2$. **E** points away from the segment along the bisector of the angle the segment subtends at P.

Set-up. Put the segment on the x-axis from $-L$ to L and the field point at $P = (x_0, z)$: perpendicular distance z from the line, at axial position x_0. Measure angles α from the perpendicular PF dropped from P to the line, positive towards $+x$ and negative towards $-x$. The end at $-L$ is seen at angle α_1 and the end at $+L$ at angle α_2. In the figure the left end is to the left of F and the right end to the right, so α_1 is negative and α_2 positive. (If P is beyond one end of the segment, both ends are on the same side of F and both angles have the same sign; everything below still holds, because α_1 is always the more negative of the two.)

Step 1: the element. An element $\lambda\,\mathrm{d}x'$ at $(x', 0)$ is joined to P by $\mathbf{r} = (x_0 - x', z)$, of length $R = \sqrt{(x_0 - x')^2 + z^2}$. Coulomb's law gives the components parallel and perpendicular to the line:

$$\mathrm{d}E_\parallel = \frac{\lambda}{4\pi\varepsilon_0}\frac{(x_0 - x')\,\mathrm{d}x'}{R^3}, \qquad \mathrm{d}E_\perp = \frac{\lambda}{4\pi\varepsilon_0}\frac{z\,\mathrm{d}x'}{R^3}.$$

Step 2: change variable to the angle. Write $x' - x_0 = z\tan\alpha$, so $\mathrm{d}x' = z\sec^2\alpha\,\mathrm{d}\alpha$, $R^2 = z^2\tan^2\alpha + z^2 = z^2\sec^2\alpha$, and therefore

$$\frac{\mathrm{d}x'}{R^3} = \frac{z\sec^2\alpha\,\mathrm{d}\alpha}{z^3\sec^3\alpha} = \frac{\cos\alpha\,\mathrm{d}\alpha}{z^2}.$$

Substituting, and using $x_0 - x' = -z\tan\alpha$:

$$\mathrm{d}E_\perp = \frac{\lambda}{4\pi\varepsilon_0}\,z\,\frac{\cos\alpha\,\mathrm{d}\alpha}{z^2} = \frac{\lambda}{4\pi\varepsilon_0 z}\cos\alpha\,\mathrm{d}\alpha,$$

$$\mathrm{d}E_\parallel = \frac{\lambda}{4\pi\varepsilon_0}(-z\tan\alpha)\frac{\cos\alpha\,\mathrm{d}\alpha}{z^2} = -\frac{\lambda}{4\pi\varepsilon_0 z}\sin\alpha\,\mathrm{d}\alpha.$$

Step 3: integrate between the two ends. $\int\cos\alpha\,\mathrm{d}\alpha = \sin\alpha$ and $\int(-\sin\alpha)\,\mathrm{d}\alpha = \cos\alpha$, so

$$\boxed{E_\perp = \frac{\lambda}{4\pi\varepsilon_0 z}\,(\sin\alpha_2 - \sin\alpha_1), \qquad E_\parallel = \frac{\lambda}{4\pi\varepsilon_0 z}\,(\cos\alpha_2 - \cos\alpha_1)}$$

Check. On the bisector $\alpha_2 = -\alpha_1 = \alpha$ with $\sin\alpha = L/\sqrt{L^2+z^2}$: $E_\parallel = \cos\alpha - \cos(-\alpha) = 0$ and $E_\perp = 2\lambda\sin\alpha/(4\pi\varepsilon_0 z) = 2\lambda L/[4\pi\varepsilon_0 z\sqrt{L^2+z^2}]$, which is Problem 1(a).

Optional remark: which way E points. This is not needed for the answer above. Apply the sum-to-product identities with $\bar\alpha = \frac{1}{2}(\alpha_1+\alpha_2)$ and $\Delta = \frac{1}{2}(\alpha_2-\alpha_1)$:

$$\sin\alpha_2 - \sin\alpha_1 = 2\cos\bar\alpha\sin\Delta, \qquad \cos\alpha_2 - \cos\alpha_1 = -2\sin\bar\alpha\sin\Delta.$$

So $E_\parallel/E_\perp = -\tan\bar\alpha$, and the magnitude is $E = \sqrt{E_\perp^2+E_\parallel^2} = \dfrac{\lambda}{2\pi\varepsilon_0 z}\sin\Delta$. The direction makes angle $\bar\alpha$ with the perpendicular, pointing away from the line: **E** always points along the *bisector of the angle* that the segment subtends at the field point.

1.2 Charged ring on its axis

Show that the field of a uniformly charged ring on its axis is obtained by the same method, and compare its far-field correction with that of the segment.

Step 1: the field. Take a ring of radius R and total charge Q in the xy-plane, and a point on its axis at height z. Every element $\mathrm{d}q$ is at the same distance $\sqrt{R^2+z^2}$. Elements on opposite sides of the ring have equal and opposite components perpendicular to the axis, which cancel; only the axial components survive, each a fraction $\cos\alpha = z/\sqrt{R^2+z^2}$ of the element's field:

$$E_z = \frac{1}{4\pi\varepsilon_0}\oint \frac{\mathrm{d}q}{R^2+z^2}\,\frac{z}{\sqrt{R^2+z^2}} = \frac{z}{4\pi\varepsilon_0(R^2+z^2)^{3/2}}\oint \mathrm{d}q = \frac{Qz}{4\pi\varepsilon_0(R^2+z^2)^{3/2}}.$$

Step 2: far field. Factor out z^2: $(R^2+z^2)^{3/2} = z^3(1+R^2/z^2)^{3/2}$. For $z \gg R$, $(1+u)^{-3/2} \approx 1 - \frac{3}{2}u$ with $u = R^2/z^2$:

$$E_z = \frac{Q}{4\pi\varepsilon_0 z^2}\left(1+\frac{R^2}{z^2}\right)^{-3/2} \approx \frac{Q}{4\pi\varepsilon_0 z^2}\left(1-\frac{3R^2}{2z^2}\right).$$

Comparison. Both corrections are of relative order $(\text{size})^2/z^2$ and negative, because neither distribution has a dipole moment: the first correction comes from the quadrupole. For the same size ($R = L$) the ring's correction ($\frac{3}{2}$) is three times that of the segment ($\frac{1}{2}$), because *all* the ring's charge sits at the maximum distance from the centre, whereas the segment's charge is spread from 0 to L.

1.3 Near the end of a finite segment

Sketch the field lines near the end of a finite segment: where does the infinite-line approximation fail?

Sketch (Figure 2). Near the middle and close to the line, field lines leave perpendicular to it, as for an infinite line. Towards the ends they tilt outwards, and beyond the ends they fan out radially, so that far away the pattern is that of a point charge.

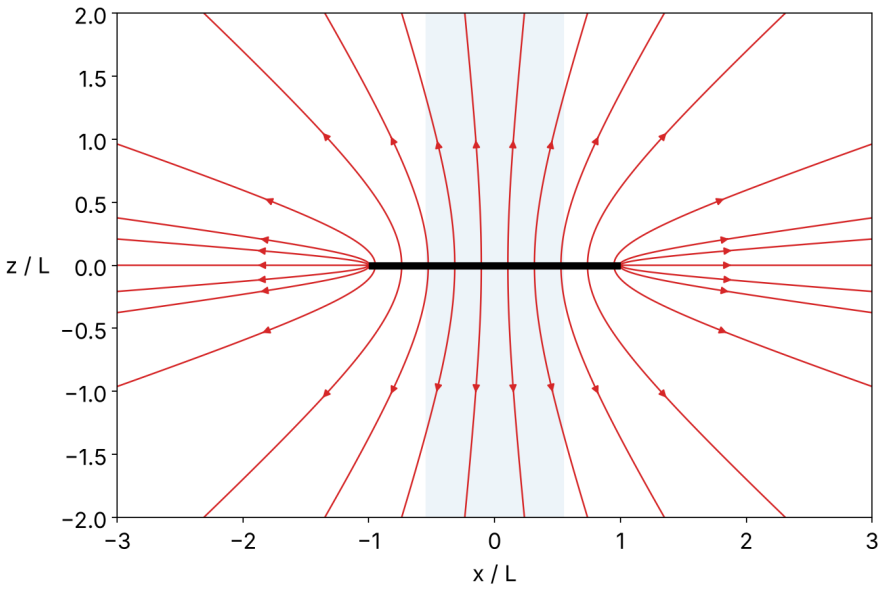


Figure 2: Field lines of a uniformly charged segment of length $2L$. In the shaded band the field is close to that of an infinite line.

Quantitative criterion. Use 1.1 at distance s from the line, a distance d inside one end (the right end, say). The other end is far away, so $\alpha_1 \to -\pi/2$ and $\sin\alpha_1 \to -1$; the near end is seen at $\tan\alpha_2 = d/s$, so $\sin\alpha_2 = d/\sqrt{d^2+s^2}$. Then

$$E_\perp = \frac{\lambda}{4\pi\varepsilon_0 s}\left(\sin\alpha_2 + 1\right), \qquad \text{compared with the infinite-line value } \frac{2\lambda}{4\pi\varepsilon_0 s}.$$

The fractional shortfall is

$$\frac{2-(1+\sin\alpha_2)}{2} = \tfrac{1}{2}(1-\sin\alpha_2) = \tfrac{1}{2}\left(1-\frac{d}{\sqrt{d^2+s^2}}\right) \approx \frac{s^2}{4d^2} \quad (d \gg s),$$

using $d/\sqrt{d^2+s^2} = (1+s^2/d^2)^{-1/2} \approx 1 - s^2/(2d^2)$. Exactly at the end ($d = 0$) the field is only half the infinite-line value. The approximation is good to 1% once $s^2/4d^2 \le 0.01$, i.e. $d \gtrsim 5s$: *it fails within a few times the distance s of either end.*

Problem 2 — Charged sphere, the Earth, and a capacitor

What you need to know for these extensions

- For a conducting sphere at potential V: $Q = 4\pi\varepsilon_0 RV$ and surface field $E = Q/(4\pi\varepsilon_0 R^2) = V/R$ (Problem 2(a)).
- **Current and current density:** total current $I = jA$ through area A; Ohm's law $j = \sigma E$ (used in 2.2), and the dielectric relaxation time of Problem 15.
- **Additional knowledge for 2.3:** the fringing formula quoted there comes from conformal mapping (complex-variable methods), which is beyond this course; it is used as a quoted result.

2.1 Smallest sphere that can hold 5 kV

What radius of sphere would be needed to hold 5 kV *before air breaks down at* $3\,\mathrm{MV\,m^{-1}}$*? Relate this to why high-voltage terminals are large and smooth.*

For a sphere at potential V, the surface field is $E = V/R$: for fixed V, a *smaller* sphere has a *larger* surface field. Requiring $E \le E_b$:

$$\frac{V}{R} \le E_b \quad \Longrightarrow \quad R \ge \frac{V}{E_b} = \frac{5\times10^3\,\mathrm{V}}{3\times10^6\,\mathrm{V\,m^{-1}}} = 1.7\,\mathrm{mm}.$$

A Van de Graaff generator at 1 MV needs $R \ge 0.33\,\mathrm{m}$. Any sharp point or edge acts locally like a sphere of small radius of curvature, so $E \approx V/R_{\mathrm{local}}$ is large there and breakdown (corona) starts there first. High-voltage electrodes are therefore large and smooth; lightning conductors are deliberately pointed for the opposite reason.

2.2 How fast would the Earth discharge?

Use the Earth's charge from (b) and a fair-weather current density of $\sim 2\times10^{-12}\,\mathrm{A\,m^{-2}}$ *to estimate how long the Earth would take to discharge without thunderstorms.*

Step 1: total current. The current flows in over the whole surface, $A = 4\pi R_\oplus^2 = 4\pi(6.4\times10^6)^2\ \mathrm{m^2} = 5.1\times10^{14}\,\mathrm{m^2}$:

$$I = jA = (2\times10^{-12})(5.1\times10^{14})\ \mathrm{A} \approx 1.0\times10^3\,\mathrm{A}.$$

Step 2: time. If the current stayed constant, the charge would be gone after

$$\tau \approx \frac{Q}{I} = \frac{4.6\times10^5\,\mathrm{C}}{1.0\times10^3\,\mathrm{A}} \approx 450\,\mathrm{s},$$

about 7–8 minutes. (In fact $I \propto Q$, so the decay is exponential with this time constant.)

Cross-check. The air's conductivity is $\sigma = j/E = (2\times10^{-12})/(100) = 2\times10^{-14}\,\mathrm{S\,m^{-1}}$. Since $Q = \varepsilon_0 EA$ (Gauss's law) and $I = \sigma EA$ (Ohm's law), $Q/I = \varepsilon_0/\sigma \approx 440\,\mathrm{s}$: the dielectric relaxation time of Problem 15. The Earth–atmosphere system is a leaky capacitor. Since the Earth plainly stays charged, something recharges it continuously: the ∼2000 thunderstorms active at any moment, each driving about 1 A (the global electric circuit).

2.3 Fringing correction to the capacitance

In (c), estimate the fractional correction to the ideal capacitance $\varepsilon_0 A/d$ *caused by fringing.*

Estimate. The extra capacitance comes from field lines that leave the edges and the outer faces. Near each edge the field spreads over a region of order d, so each edge adds roughly the capacitance of an extra strip of width $\sim (d/\pi)\times$(a logarithm). A standard result (from conformal mapping) for two strips of width w at separation d, per unit length, is

$$C \approx \frac{\varepsilon_0 w}{d}\left[1 + \frac{d}{\pi w}\left(1 + \ln\frac{2\pi w}{d}\right)\right].$$

Numbers. For $w = 5d$: $\dfrac{d}{\pi w} = \dfrac{1}{5\pi} = 0.064$ and $1 + \ln(10\pi) = 1 + 3.45 = 4.45$, so the correction is $0.064 \times 4.45 = 0.28$, a **28% increase**. The numerical solution used for the figure in Problem 2(c) gives 33% (it also includes a distant earthed boundary). Fringing is a large correction unless $w \gg d$; it falls only as $(d/w)\ln(w/d)$.

Problem 3 — Dipole field

What you need to know for these extensions

- The dipole potential and field of Problem 3, in Cartesian and polar form.
- The binomial expansion to *second* order: $(1+u)^{-1/2} \approx 1 - \frac{1}{2}u + \frac{3}{8}u^2$.
- **Field lines:** a curve everywhere tangent to $\mathbf{E}$. In polar coordinates a small step along the line has components $(\,\mathrm{d}r,\, r\,\mathrm{d}\theta)$, which must be in the ratio $E_r : E_\theta$.
- Partial derivatives with the chain rule: $\partial(r^{-n})/\partial x = -n x\, r^{-n-2}$ (since $\partial r/\partial x = x/r$).

3.1 Linear quadrupole

Compute the quadrupole potential of a linear arrangement $+q$, $-2q$, $+q$ and compare its fall-off with the dipole.

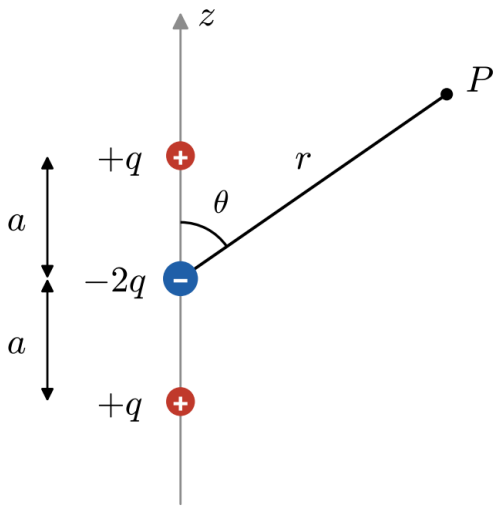


Figure 3: The linear quadrupole: $+q$, $-2q$, $+q$ spaced by a along z; P at distance r and polar angle θ.

Set-up. Charges $+q$ at $z = \pm a$ and $-2q$ at the origin; the field point P is at distance r and polar angle θ, so $z = r\cos\theta$.

Step 1: expand $1/r_\pm$ to second order. By the cosine rule, the distances from $z = \pm a$ are

$$r_\pm^2 = r^2 \mp 2ar\cos\theta + a^2 = r^2(1 + u_\pm), \qquad u_\pm = \mp\frac{2a}{r}\cos\theta + \frac{a^2}{r^2}.$$

Using $(1+u)^{-1/2} \approx 1 - \frac{1}{2}u + \frac{3}{8}u^2$ and keeping terms up to $(a/r)^2$ (in u^2 only the square of the first term is that large, $4a^2\cos^2\theta/r^2$):

$$\frac{1}{r_\pm} \approx \frac{1}{r}\left[1 \pm \frac{a}{r}\cos\theta - \frac{a^2}{2r^2} + \frac{3}{8}\cdot\frac{4a^2\cos^2\theta}{r^2}\right] = \frac{1}{r}\left[1 \pm \frac{a}{r}\cos\theta + \frac{a^2}{r^2}\cdot\frac{3\cos^2\theta - 1}{2}\right].$$

Step 2: add the three potentials.

$$V = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_+} + \frac{1}{r_-} - \frac{2}{r}\right).$$

The $1/r$ terms cancel ($1+1-2=0$: no monopole) and the $\pm(a/r)\cos\theta$ terms cancel (no dipole), leaving twice the second-order term:

$$V \approx \frac{q}{4\pi\varepsilon_0}\cdot\frac{2}{r}\cdot\frac{a^2}{r^2}\cdot\frac{3\cos^2\theta-1}{2} = \frac{qa^2(3\cos^2\theta-1)}{4\pi\varepsilon_0 r^3}.$$

(The angular factor $\frac{1}{2}(3\cos^2\theta-1)$ is the Legendre polynomial $P_2(\cos\theta)$.)

Comparison. A dipole potential falls as $1/r^2$ (field $1/r^3$); the quadrupole falls as $1/r^3$ (field $1/r^4$). Each extra order of cancellation costs one more power of a/r.

3.2 Dipole field lines

Find the equation of the dipole field lines in polar coordinates ($r \propto \sin^2\theta$) and sketch them.

Step 1: the differential equation. Along a field line, the displacement $(\mathrm{d}r,\ r\,\mathrm{d}\theta)$ is parallel to $(E_r,\ E_\theta)$, so

$$\frac{\mathrm{d}r}{r\,\mathrm{d}\theta} = \frac{E_r}{E_\theta} = \frac{2p\cos\theta/(4\pi\varepsilon_0 r^3)}{p\sin\theta/(4\pi\varepsilon_0 r^3)} = \frac{2\cos\theta}{\sin\theta}.$$

Step 2: separate and integrate.

$$\int\frac{\mathrm{d}r}{r} = \int\frac{2\cos\theta}{\sin\theta}\,\mathrm{d}\theta \quad\Longrightarrow\quad \ln r = 2\ln\sin\theta + \text{const} \quad\Longrightarrow\quad r = r_0\sin^2\theta,$$

using $\mathrm{d}(\ln\sin\theta)/\,\mathrm{d}\theta = \cos\theta/\sin\theta$. Each line is labelled by r_0, its distance from the dipole where it crosses the equatorial plane ($\theta = 90°$); all lines start and end at the origin ($\theta = 0, \pi$), as in Figure 4.

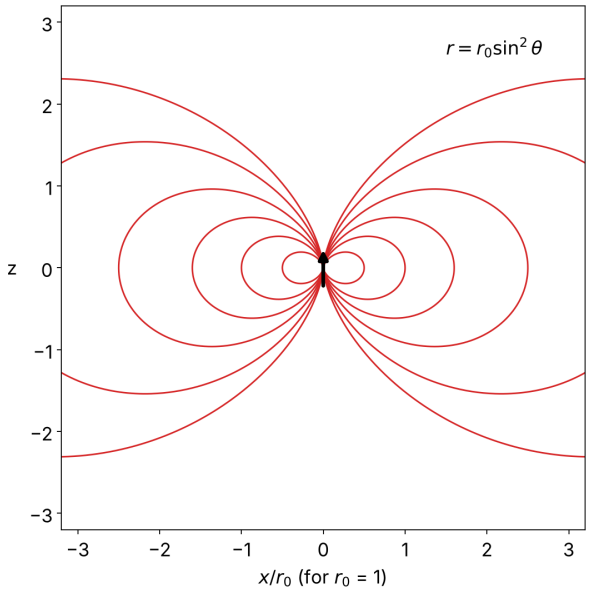


Figure 4: Field lines $r = r_0\sin^2\theta$ of a dipole at the origin pointing along $+z$ (black arrow).

3.3 Divergence and curl

Show directly that $\nabla\cdot\mathbf{E} = 0$ and $\nabla\times\mathbf{E} = 0$ for the field found here.

Write $\mathbf{E} = K(3xz,\ 3yz,\ 3z^2 - r^2)\,r^{-5}$ with $K = p/4\pi\varepsilon_0$, and use $\partial r^{-n}/\partial x = -nx\,r^{-n-2}$ (likewise for y, z).

Divergence. Each term needs the product rule:

$$\begin{aligned}\partial_x(3xz\,r^{-5}) &= 3z\,r^{-5} + 3xz(-5x\,r^{-7}) = 3z\,r^{-5} - 15x^2z\,r^{-7},\\ \partial_y(3yz\,r^{-5}) &= 3z\,r^{-5} - 15y^2z\,r^{-7},\\ \partial_z\big((3z^2-r^2)r^{-5}\big) &= (6z-2z)r^{-5} + (3z^2-r^2)(-5z\,r^{-7})\\ &= 4z\,r^{-5} - 15z^3r^{-7} + 5z\,r^{-5} = 9z\,r^{-5} - 15z^3r^{-7}.\end{aligned}$$

(In the last line $\partial_z(r^2) = 2z$.) Adding: $15z\,r^{-5} - 15z(x^2+y^2+z^2)r^{-7} = 15z\,r^{-5} - 15z\,r^{-5} = 0$. ✓

Curl, x-component.

$$\begin{aligned}\partial_y E_z &= K\big[-2y\,r^{-5} + (3z^2-r^2)(-5y\,r^{-7})\big]\\ &= K\big[-2y\,r^{-5} - 15yz^2r^{-7} + 5y\,r^{-5}\big] = K\big[3y\,r^{-5} - 15yz^2r^{-7}\big],\\ \partial_z E_y &= K\big[3y\,r^{-5} + 3yz(-5z\,r^{-7})\big]\\ &= K\big[3y\,r^{-5} - 15yz^2r^{-7}\big].\end{aligned}$$

So $(\nabla\times\mathbf{E})_x = \partial_yE_z - \partial_zE_y = 0$. The y-component follows by swapping $x \leftrightarrow y$, and $(\nabla\times\mathbf{E})_z = \partial_xE_y - \partial_yE_x = K[3yz(-5x) - 3xz(-5y)]r^{-7} = 0$. Both results must hold: $\mathbf{E} = -\nabla V$ is automatically curl-free, and the region away from the dipole contains no charge.

Problem 4 — Two dipoles

What you need to know for these extensions

- The energy $U(\theta_1,\theta_2) = -K\big[3\cos(\theta_1+\theta_2) + \cos(\theta_1-\theta_2)\big]$ with $K = p^2/(8\pi\varepsilon_0 d^3)$, from Problem 4.
- **Equilibria of a function of two variables:** both first partial derivatives vanish; a minimum has both curvatures positive, a maximum both negative, a saddle one of each. A linear, invertible change of variables does not change the classification.
- **Torque from energy:** the torque tending to increase an angle θ is $-\partial U/\partial\theta$.
- Boltzmann's constant $k_B = 1.38\times10^{-23}\,\mathrm{J\,K^{-1}}$ (for 4.3).

4.1 Equilibria of two free dipoles

Find and classify all equilibrium orientations when both dipoles are free to rotate (minima, maxima, saddle points).

Step 1: new variables. With $s = \theta_1+\theta_2$ and $\delta = \theta_1-\theta_2$ the energy separates: $U = -K(3\cos s + \cos\delta)$.

Step 2: equilibrium. By the chain rule, $\partial s/\partial\theta_1 = \partial\delta/\partial\theta_1 = 1$ and $\partial s/\partial\theta_2 = 1$, $\partial\delta/\partial\theta_2 = -1$:

$$\frac{\partial U}{\partial\theta_1} = K(3\sin s + \sin\delta) = 0, \qquad \frac{\partial U}{\partial\theta_2} = K(3\sin s - \sin\delta) = 0.$$

Adding and subtracting: $\sin s = 0$ and $\sin\delta = 0$, so $s,\delta \in \{0,\pi\}$.

Step 3: classify. In the variables (s,δ) there is no cross term, and the curvatures are $\partial^2U/\partial s^2 = 3K\cos s$ and $\partial^2U/\partial\delta^2 = K\cos\delta$. Converting back with $\theta_1 = (s+\delta)/2$, $\theta_2 = (s-\delta)/2$:

(s,δ)	(θ_1,θ_2)	Configuration	U/K	curvatures	Type
$(0,0)$	$(0,0)$	head-to-tail along the line	-4	$+,+$	minimum
$(0,\pi)$	$(\frac{\pi}{2},-\frac{\pi}{2})$	side by side, antiparallel	-2	$+,-$	saddle
$(\pi,0)$	$(\frac{\pi}{2},\frac{\pi}{2})$	side by side, parallel	$+2$	$-,+$	saddle
(π,π)	$(\pi,0)$	head-to-head along the line	$+4$	$-,-$	maximum

Only the head-to-tail state is stable against every small rotation. The antiparallel side-by-side state is stable against rotating both dipoles the same way (s) but unstable against counter-rotation (δ).

4.2 Torques and angular momentum

Find the torque on dipole 2 as a function of θ_1 and θ_2, and show that the torques on the two dipoles are not equal and opposite. Where does the missing angular momentum go?

Step 1: torques. The torque tending to increase θ_i is $\tau_i = -\partial U/\partial\theta_i$, so from 4.1, Step 2,

$$\tau_1 = -K\big[3\sin(\theta_1+\theta_2)+\sin(\theta_1-\theta_2)\big], \qquad \tau_2 = -K\big[3\sin(\theta_1+\theta_2)-\sin(\theta_1-\theta_2)\big].$$

Step 2. $\tau_1+\tau_2 = -6K\sin(\theta_1+\theta_2)$, which is generally non-zero: the torques are not equal and opposite.

Step 3: where it goes. Rotating the *whole* system rigidly (both dipoles and the line joining them together) leaves both θ_i, and hence U, unchanged, since the angles are measured from that line. So the total torque on the system, *including* the moment of the forces about a common point, must vanish. The forces between dipoles are not along the line joining them (they are non-central), so the force on dipole 2 and the equal and opposite force on dipole 1 have perpendicular components $F_\perp$ that form a couple of moment $dF_\perp$. This couple supplies exactly the balance, $dF_\perp = 6K\sin(\theta_1+\theta_2)$. If the dipoles were free, the "missing" angular momentum would appear as orbital angular momentum of the pair; with the dipoles on fixed pivots, it is taken up by the supports.

4.3 Two water molecules

Estimate the interaction energy of two water molecules ($p \approx 6.2\times10^{-30}\,\mathrm{C\,m}$) at 0.3 nm and compare with k_BT at room temperature.

For the head-to-tail minimum, $U = -4K = -2p^2/(4\pi\varepsilon_0 r^3)$:

$$|U| = \frac{2(8.99\times10^{9})(6.2\times10^{-30})^2}{(3\times10^{-10})^3}\,\mathrm{J} = \frac{6.9\times10^{-49}}{2.7\times10^{-29}}\,\mathrm{J} = 2.6\times10^{-20}\,\mathrm{J} \approx 0.16\,\mathrm{eV},$$

$$\frac{|U|}{k_BT} = \frac{2.6\times10^{-20}}{(1.38\times10^{-23})(300)} \approx 6.$$

The dipole–dipole energy is several times k_BT, so thermal motion cannot randomise neighbouring water molecules: water is a strongly associated liquid. (At 0.3 nm the point-dipole model is only rough; the real hydrogen bond, ≈ 0.2 eV, is of the same order.)

Problem 5 — Self-energy

What you need to know for these extensions

- The three equivalent forms of electrostatic energy from Problem 5: $\frac{1}{2}\int \rho V \,\mathrm{d}\tau$, $\frac{\varepsilon_0}{2}\int E^2 \,\mathrm{d}\tau$, and building up the charge piece by piece.
- **Additional knowledge for 5.2 (nuclear physics):** mirror nuclei have the numbers of protons and neutrons swapped; the strong force is (nearly) the same for protons and neutrons, so their energies differ mainly by the Coulomb energy. Nuclear radii $R \approx 1.2A^{1/3}$ fm, and $e^2/(4\pi\varepsilon_0) = 1.44\,\mathrm{MeV\,fm}$.
- **Integration by parts in 3D** (for 5.3): $\int \nabla V_1 \cdot \nabla V_2 \,\mathrm{d}\tau = -\int V_2 \nabla^2 V_1 \,\mathrm{d}\tau$ when the surface term at infinity vanishes.

5.1 A charged spherical shell

Repeat for a charged spherical shell and explain why $3/5$ *becomes* $1/2$.

Step 1: field energy. For a shell of radius a and charge Q, Gauss's law gives $E = 0$ inside and $E = Q/(4\pi\varepsilon_0 r^2)$ outside. The field energy is

$$U = \frac{\varepsilon_0}{2}\int_a^\infty \left(\frac{Q}{4\pi\varepsilon_0 r^2}\right)^2 4\pi r^2 \,\mathrm{d}r = \frac{Q^2}{8\pi\varepsilon_0}\int_a^\infty \frac{\mathrm{d}r}{r^2} = \frac{Q^2}{8\pi\varepsilon_0 a} = \frac{1}{2}\,\frac{Q^2}{4\pi\varepsilon_0 a}.$$

(Equivalently $U = \frac{1}{2}\int \sigma V \,\mathrm{d}A = \frac{1}{2}QV$ with $V = Q/4\pi\varepsilon_0 a$ on the shell.)

Why $\frac{1}{2}$ instead of $\frac{3}{5}$. The field outside is the same for both. The uniform ball has an additional interior field that the shell lacks; Problem 5 found it contributes $Q^2/(40\pi\varepsilon_0 a) = \frac{1}{10}\,Q^2/(4\pi\varepsilon_0 a)$. And $\frac{1}{2} + \frac{1}{10} = \frac{3}{5}$. Equivalently, for a given radius the charges in a ball are on average closer together. A shell is the minimum-energy arrangement, which is why charge on a conductor goes to its surface.

5.2 Mirror nuclei

Use the result to estimate the electrostatic contribution to the mass difference between mirror nuclei (e.g. ^{3}*He and* ^{3}*H) and compare with experiment.*

Step 1: model. Treat the nucleus as a uniformly charged sphere of radius R containing Z protons. Problem 5's formula with $Q = Ze$ gives $\frac{3}{5}(Ze)^2/(4\pi\varepsilon_0 R)$, but this includes the (unphysical) self-energy of each proton. There are $Z(Z-1)/2$ distinct proton pairs, not $Z^2/2$, so the Coulomb energy between protons is

$$U_C = \tfrac{3}{5}\,Z(Z-1)\,\frac{e^2}{4\pi\varepsilon_0 R}.$$

Step 2: difference. ^{3}He ($Z = 2$) has $Z(Z-1) = 2$; ^{3}H ($Z = 1$) has 0:

$$\Delta U_C = \frac{3}{5}\times 2 \times \frac{e^2}{4\pi\varepsilon_0 R} = \frac{1.2 \times 1.44\,\mathrm{MeV\,fm}}{R}.$$

Step 3: numbers. With $R = 1.2A^{1/3}$ fm $= 1.2\times 3^{1/3} = 1.73$ fm, $\Delta U_C \approx 1.0$ MeV. Alternatively use the measured charge radius of ^{3}He, $r_{\mathrm{rms}} = 1.97$ fm. For a uniform sphere, $\langle r^2\rangle = \int_0^R r^2 \cdot 4\pi r^2\,\mathrm{d}r/\frac{4}{3}\pi R^3 = \frac{3}{5}R^2$, so the equivalent $R = \sqrt{5/3}\,r_{\mathrm{rms}} = 2.54$ fm, giving 0.68 MeV.

Experiment. The binding energies are 8.48 MeV (^{3}H) and 7.72 MeV (^{3}He), differing by 0.76 MeV. The simple electrostatic model gets the size right; historically this was one of the ways nuclear radii were first measured.

5.3 Why point charges have infinite self-energy

Explain why the energy of a point charge diverges, while the energy of a pair of point charges, excluding self-energies, is finite.

Self-energy. Near a point charge $E^2 \propto 1/r^4$, and the energy in a shell from r to $r + \mathrm{d}r$ is $\propto r^{-4} \cdot 4\pi r^2 \,\mathrm{d}r \propto \mathrm{d}r/r^2$. Integrating from a small radius a outwards gives $\int_a^\infty \mathrm{d}r/r^2 = 1/a$, which diverges as $a \to 0$ (the 5.1 result, $Q^2/8\pi\varepsilon_0 a$, with $a \to 0$).

Pair. The total field energy of two charges is

$$\frac{\varepsilon_0}{2}\int (\mathbf{E}_1 + \mathbf{E}_2)^2 \,\mathrm{d}\tau = \underbrace{\frac{\varepsilon_0}{2}\int E_1^2 \,\mathrm{d}\tau}_{\infty} + \underbrace{\frac{\varepsilon_0}{2}\int E_2^2 \,\mathrm{d}\tau}_{\infty} + \underbrace{\varepsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 \,\mathrm{d}\tau}_{\text{cross term}}.$$

The cross term is finite. With $\mathbf{E}_i = -\nabla V_i$ and integration by parts, $\varepsilon_0 \int \nabla V_1 \cdot \nabla V_2 \,\mathrm{d}\tau = -\varepsilon_0 \int V_2 \nabla^2 V_1 \,\mathrm{d}\tau = \int V_2 \rho_1 \,\mathrm{d}\tau = q_1 V_2(\mathbf{r}_1) = \dfrac{q_1 q_2}{4\pi\varepsilon_0 r_{12}}$, using Poisson's equation $\nabla^2 V_1 = -\rho_1/\varepsilon_0$. Near each charge its integrand behaves only as $r^{-2} \cdot r^2$, which is integrable.

The infinite self-terms do not depend on where the charges are, so they never affect forces or energy *differences*. That is why the usual pair energy $\frac{1}{2}\sum_{i \neq j} q_i q_j / 4\pi\varepsilon_0 r_{ij}$ simply omits them.

Problem 6 — Conductors

What you need to know for these extensions

- **Uniqueness theorem:** in a region where $\nabla^2 V = 0$, the solution is uniquely fixed by the values of V on the boundary. (If V_1 and V_2 both work, $W = V_1 - V_2$ is zero on the boundary and satisfies Laplace's equation, and $\int |\nabla W|^2 \,\mathrm{d}\tau = 0$ forces $W = 0$.)
- Problem 6(c): the charge on the inner and outer surfaces of a conductor surrounding a charge.
- **For 6.3:** the skin depth $\delta = \sqrt{2/(\mu_r \mu_0 \sigma \omega)}$, with μ_r the relative permeability of the metal (Sheet 2, Problems 36–38): a.c. currents and fields penetrate a conductor only to about δ.

6.1 Uniqueness and the field-free cavity

Use the uniqueness theorem to show that the field inside an empty cavity of any shape in a conductor is zero.

Step 1: boundary values. The cavity wall is part of the conductor, which is an equipotential, so $V = V_0$ everywhere on the wall (whatever charges are outside, V_0 is some constant).

Step 2: a candidate solution. Inside the empty cavity there is no charge, so $\nabla^2 V = 0$. The constant function $V = V_0$ satisfies Laplace's equation (all its derivatives vanish) and matches the boundary values.

Step 3: uniqueness. The solution of Laplace's equation in a region is fixed by its values on the boundary, so $V = V_0$ is *the* solution, and $\mathbf{E} = -\nabla V = 0$ throughout the cavity, for any shape.

This replaces the field-line argument of Problem 6(c).

6.2 Off-centre charge in a spherical shell

A point charge sits off-centre inside a neutral conducting spherical shell. Sketch the induced charge on the inner and outer surfaces and the field outside.

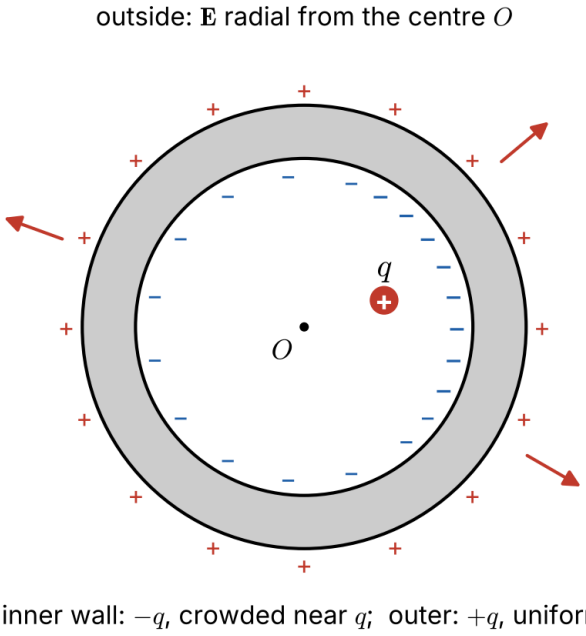


Figure 5: An off-centre charge q in a neutral conducting shell: the induced $-q$ on the inner wall crowds towards q, while the $+q$ on the outer surface is uniform. Outside, the field is radial from the shell centre O.

- **Inner surface:** total charge $-q$ (Gauss's law on a surface within the metal), distributed *non-uniformly*: densest on the part of the wall nearest the charge, so that the field of q is cancelled throughout the metal.
- **Outer surface:** total $+q$ (the shell is neutral), distributed *uniformly*. The field in the metal is zero, so the outer charge "sees" nothing of the interior arrangement; the only arrangement that makes the outer surface an equipotential with zero field just inside is the uniform one (by uniqueness, applied to the region outside).
- **Outside:** $\mathbf{E} = q\,\hat{\mathbf{r}}/(4\pi\varepsilon_0 r^2)$ measured from the *centre of the shell*, not from the charge. An observer outside cannot tell where inside the charge sits.

6.3 Cars and aircraft in lightning

Why does a car or aircraft protect its occupants from lightning even though it is not a closed conductor?

- The lightning current (tens of kA) follows the lowest-resistance path, which is the metal skin. Current in a conducting shell flows on its outside, and the field it produces inside is small.
- A strike lasts tens of microseconds, so its frequency content is $\sim$10 kHz to 100 kHz. The skin depth in a metal of conductivity σ and relative permeability μ_r, at angular frequency

ω, is

$$\delta = \sqrt{\frac{2}{\mu_r \mu_0 \sigma \omega}}.$$

Aircraft (aluminium). Aluminium is non-magnetic ($\mu_r = 1 + 2 \times 10^{-5}$), so its permeability is μ_0 to within two parts in 10^5. With $\sigma = 3.8 \times 10^7\,\mathrm{S\,m^{-1}}$ and $\omega = 2\pi \times 50\,\mathrm{kHz} = 3.1 \times 10^5\,\mathrm{s^{-1}}$,

$$\delta = \sqrt{\frac{2}{(4\pi \times 10^{-7})(3.8 \times 10^7)(3.1 \times 10^5)}}\ \mathrm{m} \approx 0.37\,\mathrm{mm}.$$

Cars (mostly steel). Mild steel is ferromagnetic, with μ_r of order a few hundred, and $\sigma \approx 6 \times 10^6\,\mathrm{S\,m^{-1}}$. Taking $\mu_r \approx 100$ as a conservative value gives $\delta \approx 0.09\,\mathrm{mm}$, about four times thinner than in aluminium. (The very large lightning current can partly saturate the steel and lower μ_r, so this value is only indicative, but it does not change the conclusion.)

In both cases the skin depth is well below the thickness of the panels (1 mm to 2 mm for aircraft skin, 0.7 mm to 1 mm for car body steel), so the current flows on the outer surface. The windows are small compared with the wavelength (kilometres), so the shell acts as an almost closed Faraday cage.

- It is the metal shell, *not* the rubber tyres, that protects: a spark that has crossed kilometres of air easily crosses a few centimetres of rubber. Convertibles and cars with plastic bodies give little protection.

Problem 7 — Conducting sphere between two charges

What you need to know for these extensions

- **Image in a sphere:** a charge Q at distance D from the centre of an *earthed* conducting sphere of radius a has image $-Qa/D$ at distance a^2/D from the centre, on the same line. (Check: every point of the sphere is at distances from Q and the image in the ratio D/a, so the two potentials cancel there.) For an isolated neutral sphere one adds $+Qa/D$ at the centre.
- The binomial expansion $(1 \mp \varepsilon)^{-2} = 1 \pm 2\varepsilon + 3\varepsilon^2 \pm 4\varepsilon^3 + \cdots$.
- **For 7.3: $\mathbf{D} = \epsilon\varepsilon_0\mathbf{E}$** and the boundary conditions at a dielectric surface (V continuous, normal D continuous), as in Problem 14.

7.1 The exact image solution

Use the full image solution to find the next correction, and estimate how small a/d must be for 1% accuracy.

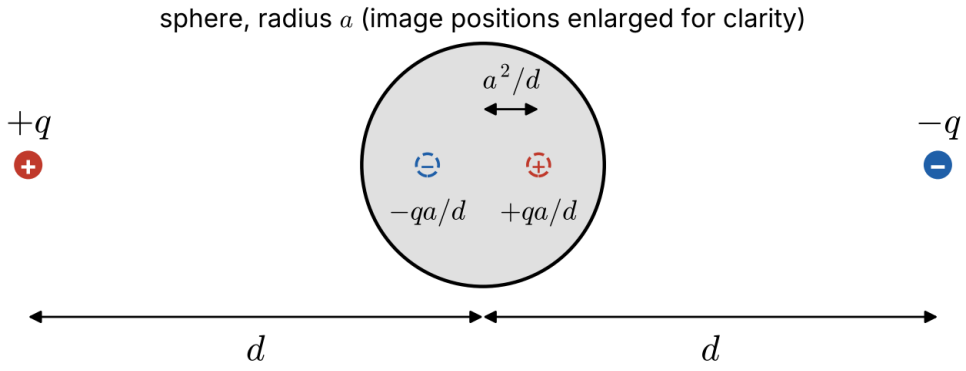


Figure 6: The exact image solution: images $\mp qa/d$ at distance a^2/d from the centre, on the line joining the charges. Image positions enlarged for clarity.

Step 1: the images. $+q$ at $x = -d$ has image $-qa/d$ at $x = -a^2/d$, and $-q$ at $x = +d$ has image $+qa/d$ at $x = +a^2/d$. The two central charges ($\pm qa/d$) that an isolated sphere would need cancel, so the images above are the complete solution: the sphere is neutral and, by symmetry, at potential zero. The solution is exact.

Step 2: exact force on $+q$ (positive towards $+x$, i.e. towards $-q$). The three other charges are: $-q$ at distance $2d$ (attractive); $-qa/d$ at distance $d - a^2/d$ (attractive); $+qa/d$ at distance $d + a^2/d$ (repulsive). With $\varepsilon = a^2/d^2$, so that $d \mp a^2/d = d(1 \mp \varepsilon)$:

$$F = \frac{q^2}{4\pi\varepsilon_0}\left[\frac{1}{(2d)^2} + \frac{a/d}{d^2(1-\varepsilon)^2} - \frac{a/d}{d^2(1+\varepsilon)^2}\right] = \frac{q^2}{4\pi\varepsilon_0 d^2}\left[\frac{1}{4} + \frac{a}{d}\Big((1-\varepsilon)^{-2} - (1+\varepsilon)^{-2}\Big)\right].$$

Step 3: expand. From the binomial series, $(1-\varepsilon)^{-2} - (1+\varepsilon)^{-2} = 4\varepsilon + 8\varepsilon^3 + \cdots$ (the even powers cancel). Dividing by $F_0 = q^2/(16\pi\varepsilon_0 d^2)$, i.e. multiplying the bracket by 4:

$$\frac{F}{F_0} = 1 + 4\frac{a}{d}(4\varepsilon + 8\varepsilon^3) = 1 + 16\left(\frac{a}{d}\right)^3 + 32\left(\frac{a}{d}\right)^7 + \cdots.$$

(In closed form, $F/F_0 = 1 + 16(a/d)^3/[1-(a/d)^4]^2$, as quoted in the main solution.)

Step 4: accuracy. The next correction is of order $(a/d)^7$, not $(a/d)^5$: by symmetry the applied field has no gradient at the centre, so no quadrupole is induced. Relative to the leading correction it is smaller by $2(a/d)^4$. The approximate formula is within 1% of the exact result as long as $32(a/d)^7 < 0.01\,[1 + 16(a/d)^3]$; trying values, this holds up to $a/d \approx 0.34$. Even at $a/d = 0.5$ the error is only $0.25/3 = 8\%$.

7.2 Earthing the sphere

> *Repeat with the sphere earthed rather than isolated: what changes, and why?*

Nothing changes. The isolated neutral sphere sits exactly midway between $+q$ and $-q$, on the plane where the potential is zero by symmetry, so it is already at earth potential. Connecting it to earth makes no charge flow. (If the sphere were off the mid-plane, its potential would not be zero, earthing would let net charge flow onto it, and the answer would change.)

7.3 A dielectric sphere

> *Replace the conductor by a dielectric sphere of relative permittivity ϵ and show the factor becomes* $1 + 16\,\frac{\epsilon-1}{\epsilon+2}\,\frac{a^3}{d^3}$.

Step 1: dielectric sphere in a uniform field. As in Problem 7 (and Problem 20), try a uniform field inside and the applied field plus a dipole outside:

$$V_{\text{in}} = -Cr\cos\theta, \qquad V_{\text{out}} = -E_0 r\cos\theta + \frac{A\cos\theta}{r^2}.$$

Step 2: boundary conditions at $r = a$ (cancelling the common $\cos\theta$).

- V continuous (equivalently, tangential **E** continuous): $-Ca = -E_0 a + A/a^2$, so $C = E_0 - A/a^3$.
- Normal **D** continuous, $D_r = -\epsilon\varepsilon_0\,\partial V/\partial r$ inside and $-\varepsilon_0\,\partial V/\partial r$ outside: $\epsilon C = E_0 + 2A/a^3$.

Substituting the first into the second:

$$\epsilon E_0 - \frac{\epsilon A}{a^3} = E_0 + \frac{2A}{a^3} \quad\Longrightarrow\quad \frac{A}{a^3}(\epsilon+2) = (\epsilon-1)E_0 \quad\Longrightarrow\quad A = \frac{\epsilon-1}{\epsilon+2}\,a^3 E_0.$$

The outside term $A\cos\theta/r^2$ is a dipole potential $p\cos\theta/(4\pi\varepsilon_0 r^2)$ with

$$p = 4\pi\varepsilon_0 A = 4\pi\varepsilon_0 a^3\,\frac{\epsilon-1}{\epsilon+2}\,E_0.$$

Step 3. The rest of Problem 7 is unchanged, with p reduced by the factor $(\epsilon-1)/(\epsilon+2)$:

$$\frac{F}{F_0} = 1 + 16\,\frac{\epsilon-1}{\epsilon+2}\,\frac{a^3}{d^3}.$$

For $\epsilon \to \infty$ the factor tends to 1, the conductor result; for $\epsilon = 1$ it vanishes and the sphere is invisible.

Problem 8 — Thundercloud

What you need to know for these extensions

- The ground field of Problem 8: each charge Q at height h with its image gives $E_z = -2kQh/(\rho^2+h^2)^{3/2}$ at the ground.
- **Surface charge on a conductor** $\sigma = \varepsilon_0 E_n$ (Problem 6(b)), with $\hat{\mathbf{n}}$ pointing out of the conductor (upwards here).
- Integration over a plane in polar coordinates: area element $2\pi\rho\,\mathrm{d}\rho$ for a ring of radius ρ.

8.1 Induced charge on the ground

Find the total induced ground charge (why is it zero here?) and sketch $\sigma(\rho)$.

Step 1: density. At the ground $\sigma = \varepsilon_0 E_z$, with E_z from Problem 8 and $k = 1/(4\pi\varepsilon_0)$:

$$\sigma(\rho) = \varepsilon_0 \cdot \left(-\frac{2}{4\pi\varepsilon_0}\right)\left[\frac{Q_1 h_1}{(\rho^2+h_1^2)^{3/2}} + \frac{Q_2 h_2}{(\rho^2+h_2^2)^{3/2}}\right] = -\frac{1}{2\pi}\left[\frac{Q_1 h_1}{(\rho^2+h_1^2)^{3/2}} + \frac{Q_2 h_2}{(\rho^2+h_2^2)^{3/2}}\right].$$

Step 2: total. For each term, substitute $w = \rho^2 + h^2$, $\mathrm{d}w = 2\rho\,\mathrm{d}\rho$:

$$\int_0^\infty \frac{h\,2\pi\rho\,\mathrm{d}\rho}{(\rho^2+h^2)^{3/2}} = \pi h\int_{h^2}^\infty w^{-3/2}\,\mathrm{d}w = \pi h\left[-2w^{-1/2}\right]_{h^2}^\infty = \pi h\cdot\frac{2}{h} = 2\pi.$$

So

$$Q_{\text{ground}} = \int_0^\infty \sigma\, 2\pi\rho\, \mathrm{d}\rho = -\frac{1}{2\pi}\left(Q_1 \cdot 2\pi + Q_2 \cdot 2\pi\right) = -(Q_1 + Q_2) = 0.$$

Why zero. The charge induced on a large conducting plane by a charge Q above it is exactly $-Q$ (it mirrors the image): every field line from Q ends on the plane. The cloud is neutral overall, so the induced charges cancel.

Sketch (Figure 7). $\sigma = \varepsilon_0 \times 12.8\,\mathrm{kV\,m^{-1}} = +113\,\mathrm{nC\,m^{-2}}$ directly below, falling through zero at 11.0 km to a weak negative tail (minimum $-1.8\,\mathrm{nC\,m^{-2}}$ at 15.7 km). The positive charge is concentrated close in; the compensating negative charge is spread thinly far out.

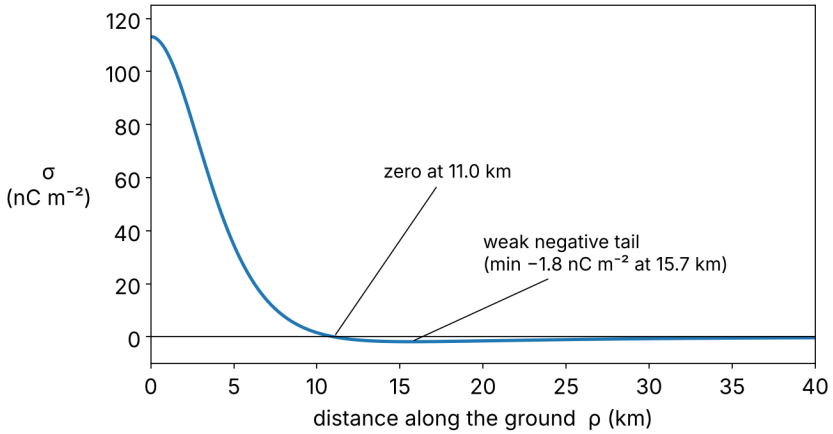


Figure 7: Surface charge density induced on the ground by the thundercloud dipole.

8.2 Potential difference across a person

Estimate the potential difference between the head and feet of a person 1.8 m *tall standing directly under the cloud.*

In the undisturbed field, the air at head height is at $V = Eh = (12.8 \times 10^3\,\mathrm{V\,m^{-1}})(1.8\,\mathrm{m}) \approx 23\,\mathrm{kV}$ relative to the ground. But a person is a reasonable conductor in contact with the ground, so the body is an *equipotential at earth potential*: there is essentially no potential difference between head and feet. Instead, the field lines that would have reached the ground crowd onto the person, concentrating the field near the head (field enhancement on a protruding conductor, as at the small sphere of extension 2.1). This is why hair stands on end and why an isolated person is a likely starting point for an upward leader.

8.3 A small positive charge at the cloud base

Real clouds also carry a small positive charge at their base. Add $+10\,\mathrm{C}$ *at* 2 km *and recompute the field directly below.*

Directly below ($\rho = 0$), each charge with its image contributes $E_z = -2kQh/h^3 = -2kQ/h^2$. The new charge adds

$$\Delta E_z = -\frac{2(8.99 \times 10^9)(10)}{(2 \times 10^3)^2}\,\mathrm{V\,m^{-1}} = -\frac{1.80 \times 10^{11}}{4 \times 10^6}\,\mathrm{V\,m^{-1}} = -45\,\mathrm{kV\,m^{-1}},$$

$$E_z = 12.8 - 44.9 = -32\,\mathrm{kV\,m^{-1}}.$$

The field reverses to $32\,\mathrm{kV\,m^{-1}}$ *downwards*. A small charge low down dominates the large charges higher up because the field of each charge-plus-image falls as $1/h^2$. Field measurements at the ground therefore mainly probe the lowest charge layer.

Problem 9 — Charge in a wedge

What you need to know for these extensions

- The image construction of Problem 9: reflection in a line at angle β maps $\phi \to 2\beta - \phi$ and changes the sign of the charge.
- The radial force from a charge q' at angular separation Δ on the circle of radius x: $F_r = \dfrac{qq'}{4\pi\varepsilon_0}\dfrac{x(1-\cos\Delta)}{|\mathbf{r}-\mathbf{r}'|^3}$, with $|\mathbf{r}-\mathbf{r}'| = 2x\sin(\Delta/2)$.
- Surface charge $\sigma = \varepsilon_0 E_n$ on a conductor (Problem 6(b)).

9.1 A 90° wedge

Repeat for a 90° wedge (three images) and compare the force at the same distance from the edge.

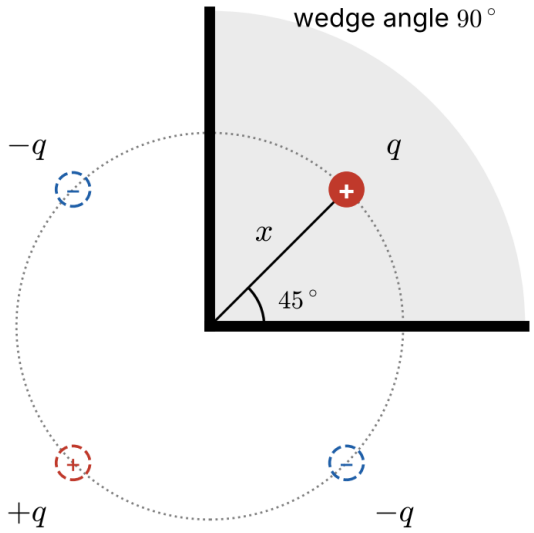


Figure 8: The 90° wedge: the real charge at 45° and its three images, all on the circle of radius x.

Images. Sheets at $\phi = 0$ and 90°; the charge is on the bisector at 45°, distance x from the edge. Reflecting in $\phi = 0$ ($\phi \to -\phi$) gives $-q$ at $-45°$; reflecting in $\phi = 90°$ ($\phi \to 180° - \phi$) gives $-q$ at 135°; reflecting either of these in the other sheet gives $+q$ at 225°. The set closes with three images ($360°/90° = 4$ charges).

Force. With $F_0 = q^2/4\pi\varepsilon_0 x^2$:

- two $-q$ at $\Delta = 90°$: distance $2x\sin 45° = \sqrt{2}x$, $1-\cos\Delta = 1$; each gives $-F_0 \cdot 1/(\sqrt{2})^3 = -F_0/(2\sqrt{2})$;

- one $+q$ at $\Delta = 180°$: distance $2x$, $1 - \cos\Delta = 2$; gives $+F_0 \cdot 2/8 = +F_0/4$.

$$F_r = F_0\left[-\frac{2}{2\sqrt{2}} + \frac{1}{4}\right] = -F_0\left(\frac{1}{\sqrt{2}} - \frac{1}{4}\right)$$
$$= -\left(\frac{4}{\sqrt{2}} - 1\right)\frac{q^2}{16\pi\varepsilon_0 x^2} = -(2\sqrt{2} - 1)\,\frac{q^2}{16\pi\varepsilon_0 x^2} = -1.83\,\frac{q^2}{16\pi\varepsilon_0 x^2}.$$

This is weaker than the 60° wedge (2.69): in the wider wedge the charge is farther from each sheet ($x\sin 45° = 0.71x$ rather than $x\sin 30° = 0.5x$).

9.2 Induced charge on one sheet

Find the induced charge density on one sheet as a function of distance from the edge.

Set-up. Use Cartesian axes with z along the edge, the sheet $\phi = 0$ as the half-plane $y = 0$, $x > 0$, and the wedge interior on the $+y$ side. Take a point on this sheet at $(s, 0, z)$: distance s from the edge and height z along it (the real charge is at height 0). Then $\sigma = \varepsilon_0 E_y$, the component of $\mathbf{E}$ normal to the sheet, pointing into the wedge.

Step 1: one charge. A charge Q at $(x\cos\phi,\ x\sin\phi,\ 0)$ is joined to the point by the vector

$$(s - x\cos\phi,\ -x\sin\phi,\ z), \qquad \text{of length } R_\phi, \quad R_\phi^2 = s^2 + x^2 - 2sx\cos\phi + z^2.$$

Its normal field component is

$$E_y = -\frac{Qx\sin\phi}{4\pi\varepsilon_0 R_\phi^3}.$$

Step 2: pair the charges. The six charges come in mirror pairs about this sheet, at $\pm\phi$ with opposite signs; both members give the same E_y (two sign changes), so each pair contributes twice the first member's. For the pairs ($+q$ at 30°, $-q$ at 330°), ($-q$ at 90°, $+q$ at 270°) and ($+q$ at 150°, $-q$ at 210°), with $\sin 30° = \sin 150° = \frac{1}{2}$ and $\sin 90° = 1$:

$$\sigma(s,z) = \varepsilon_0 E_y = \frac{qx}{4\pi}\left[-\frac{1}{R_{30}^3} + \frac{2}{R_{90}^3} - \frac{1}{R_{150}^3}\right].$$

Along $z = 0$ (Figure 9) σ is negative, peaks near the foot of the perpendicular from q ($s = x\cos 30°$), and vanishes at the edge: the three terms cancel at $s = 0$, where all $R_\phi = x$ ($-1 + 2 - 1 = 0$); fields cannot penetrate a sharp concave corner. Integrating numerically over the whole sheet gives $-q/2$; the other sheet carries the other $-q/2$, so all the field lines from q end on the sheets.

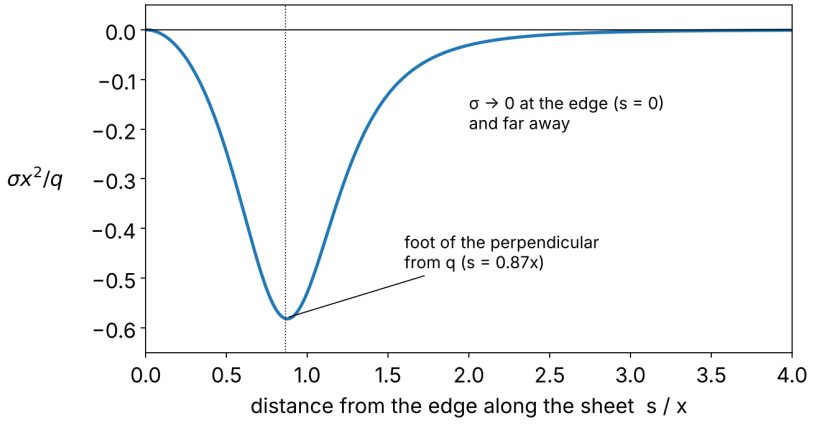


Figure 9: Induced charge density on one sheet of the 60° wedge, along the line through the foot of the perpendicular from q.

9.3 General wedge angle π/n

For a general wedge angle π/n, set up the image sum for the force and evaluate it numerically for $n = 1$ to 6.

Step 1: images. Sheets at $\phi = 0$ and $\phi = \pi/n$; the charge is on the bisector at angle $\alpha = \pi/2n$. Repeated reflection generates $2n$ charges on the circle of radius x: $+q$ at $\alpha + 2\pi k/n$ and $-q$ at $-\alpha + 2\pi k/n$, for $k = 0, \ldots, n-1$. (Two reflections, in lines at angle π/n to each other, make a rotation by $2\pi/n$, which generates the copies; one reflection, $\phi \to -\phi$, gives the opposite-sign set.)

Step 2: a neat simplification. For a charge separated by angle Δ, $|\mathbf{r} - \mathbf{r}'| = 2x\sin(\Delta/2)$ and $1 - \cos\Delta = 2\sin^2(\Delta/2)$, so

$$F_r = \frac{qq'}{4\pi\varepsilon_0}\,\frac{x \cdot 2\sin^2(\Delta/2)}{8x^3\sin^3(\Delta/2)} = \frac{qq'}{16\pi\varepsilon_0 x^2\sin(\Delta/2)}.$$

The $+q$ images are at $\Delta = 2\pi k/n$ ($k = 1, \ldots, n-1$), and the $-q$ images at $\Delta = 2\alpha + 2\pi k/n = \pi(2k+1)/n$ ($k = 0, \ldots, n-1$), so

$$F_r = \frac{q^2}{16\pi\varepsilon_0 x^2}\left[\sum_{k=1}^{n-1}\frac{1}{\sin(\pi k/n)} - \sum_{k=0}^{n-1}\frac{1}{\sin\big(\pi(2k+1)/2n\big)}\right].$$

Check, $n = 3$: first sum $2/\sin 60^\circ = 4/\sqrt{3}$; second sum $1/\sin 30^\circ + 1/\sin 90^\circ + 1/\sin 150^\circ = 5$; coefficient $4/\sqrt{3} - 5 = -2.69$, as in Problem 9.

Step 3: numbers (coefficient of $q^2/16\pi\varepsilon_0 x^2$; negative = towards the edge):

n	1	2	3	4	5	6
wedge angle	180°	90°	60°	45°	36°	30°
coefficient	−1.00	−1.83	−2.69	−3.56	−4.44	−5.32

$n = 1$ is a single plane (the classic $q^2/16\pi\varepsilon_0 x^2$), $n = 2$ reproduces 9.1 and $n = 3$ Problem 9. The coefficient grows almost linearly with n: as the wedge narrows, the charge is closer to both sheets.

Problem 10 — Connecting two spheres

What you need to know for these extensions

- The reduced circuit of Problem 10: C_1, R, C_2 in series, equivalent to $C_s = C_1C_2/(C_1 + C_2)$, with initial voltage difference $\Delta V_0 = V_2 - V_1$.
- **Inductance (from Sheet 2):** an inductor L adds a voltage $L\,\mathrm{d}I/\mathrm{d}t$ and stores energy $\frac{1}{2}LI^2$. A series LCR circuit oscillates if $R < 2\sqrt{L/C}$.
- Multiplying an equation of motion by the velocity (here $I = \dot{q}$) to obtain an energy equation.

10.1 Heat with inductance

With inductance $L \neq 0$, show that the total heat is unchanged in the underdamped case if radiation is neglected.

Step 1: circuit equation. Let q be the charge transferred from sphere 2 to sphere 1. Round the loop, the driving voltage $\Delta V_0 - q/C_s$ (Problem 10, Step 4) now drives both the resistor and the inductor:

$$L\ddot{q} + R\dot{q} = \Delta V_0 - \frac{q}{C_s}.$$

Step 2: energy equation. Multiply by $I = \dot{q}$ and recognise exact derivatives, $L\dot{q}\ddot{q} = \frac{\mathrm{d}}{\mathrm{d}t}(\frac{1}{2}L\dot{q}^2)$ and $\dot{q}\,(q/C_s - \Delta V_0) = \frac{\mathrm{d}}{\mathrm{d}t}\left(q^2/2C_s - \Delta V_0 q\right)$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\tfrac{1}{2}LI^2 + U_C(q)\right] = -I^2R, \qquad U_C(q) = U_i - \Delta V_0\, q + \frac{q^2}{2C_s}.$$

$U_C(q)$ is the electrostatic energy of the two spheres: $\dfrac{(Q_1+q)^2}{2C_1} + \dfrac{(Q_2-q)^2}{2C_2} = U_i + q(V_1 - V_2) + \dfrac{q^2}{2}\left(\dfrac{1}{C_1} + \dfrac{1}{C_2}\right)$.

Step 3: integrate from $t = 0$ (no current, $q = 0$) to $t \to \infty$ (current zero, charges settled at $q_f = C_s\Delta V_0$, where the driving voltage vanishes):

$$\int_0^\infty I^2R\,\mathrm{d}t = U_C(0) - U_C(q_f) = \Delta V_0\, C_s\Delta V_0 - \frac{(C_s\Delta V_0)^2}{2C_s} = \tfrac{1}{2}C_s\Delta V_0^2.$$

The final state is the same whatever the damping, and the inductor's energy is zero at both ends, so the total heat is the same. Underdamping only spreads the dissipation over several oscillations.

10.2 Time constant and the onset of oscillation

Estimate the time constant for a $1\,\mathrm{M\Omega}$ *wire, and the inductance below which the circuit would oscillate.*

Time constant: $\tau = RC_s = (1\times10^6\,\Omega)(2.5\,\mathrm{pF}) = 2.5\,\mu\mathrm{s}$.

Oscillation: the circuit is underdamped if $R < 2\sqrt{L/C_s}$, i.e. (squaring) if

$$L > \frac{R^2C_s}{4} = \frac{(10^6)^2(2.5\times10^{-12})}{4}\,\mathrm{H} = 0.63\,\mathrm{H}.$$

So the circuit oscillates only if L *exceeds* 0.63 H. A wire's inductance is of order microhenries, so with a megohm wire the circuit is enormously overdamped. With a low-resistance wire ($R \sim 1\,\Omega$) the condition becomes $L > 6\times10^{-13}\,\mathrm{H}$, which any wire satisfies, and it rings.

10.3 Three spheres

Repeat for three spheres connected in turn by a single wire. Is the total heat path-dependent?

Example. Three equal capacitances C at potentials V, 0, 0. Initial energy $\frac{1}{2}CV^2$. Each connection of two equal capacitors with potential difference ΔV dissipates $\frac{1}{2}C_s\Delta V^2$ with $C_s = C/2$.

- **All connected together:** charge CV shared over 3C, final $V/3$ each, energy $3\cdot\frac{1}{2}C(V/3)^2 = CV^2/6$; heat $= \frac{1}{2}CV^2 - \frac{1}{6}CV^2 = CV^2/3$.
- **1–2, then 2–3:** after 1–2, both at $V/2$, heat $\frac{1}{2}\cdot\frac{C}{2}\cdot V^2 = CV^2/4$. Then 2–3 ($V/2$ and 0) gives $V/4$ each, heat $\frac{1}{2}\cdot\frac{C}{2}\cdot(V/2)^2 = CV^2/16$. Total $5CV^2/16$, final state $(V/2, V/4, V/4)$ with energy $\frac{1}{8}CV^2 + \frac{1}{16}CV^2 = \frac{3}{16}CV^2$ (check: $\frac{1}{2} - \frac{3}{16} = \frac{5}{16}$).

The heat is always (initial energy) − (final energy). It depends only on the final state reached, not on the order of operations. Different sequences give different heat only because they end in different states; if all three are eventually connected together, the total is $CV^2/3$ whatever the route.

Problem 11 — Electrostatic pressure

What you need to know for these extensions

- **Virtual work** from Problem 11: the force in the direction of increasing x is $F_x = -(\partial U/\partial x)_Q = +(\partial U/\partial x)_V$, with $U = Q^2/2C = \frac{1}{2}CV^2$.
- **Capacitors in parallel** add: $C = C_1 + C_2$.
- **Surface tension (for 11.2):** a soap film has two surfaces, so the pressure inside a bubble of radius R exceeds that outside by $4\gamma/R$ (Laplace's formula $2\gamma/R$ per surface). For an ideal gas at constant temperature $pV = \text{const}$.
- **Additional knowledge for 11.3:** the Maxwell stress tensor is beyond the main course; its form is quoted and only its use is shown.

11.1 Force on a dielectric slab

Insert a dielectric slab of permittivity ϵ partly between the plates. Find the force pulling it in, at constant Q and at constant V, and show they agree.

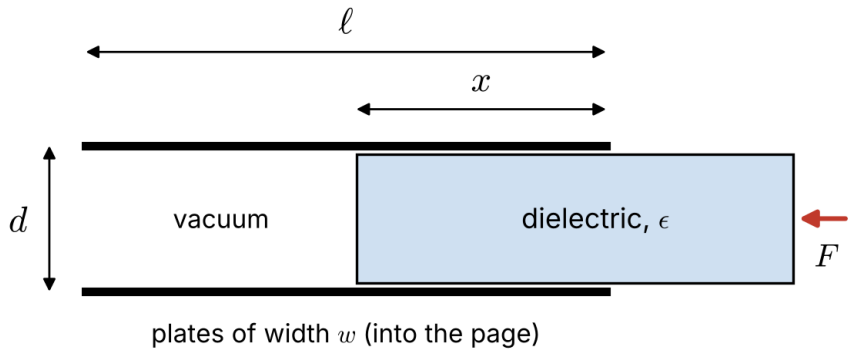


Figure 10: A dielectric slab inserted a distance x into plates of length ℓ, width w and separation d. The force F pulls the slab in.

Set-up. Plates of length ℓ (in the direction of insertion) and width w, separation d; the slab fills the gap to a depth x. The filled part (area wx) and the empty part (area $w(\ell - x)$) share the same voltage, so they are capacitors in parallel:

$$C(x) = \frac{\epsilon\varepsilon_0 wx}{d} + \frac{\varepsilon_0 w(\ell - x)}{d} = \frac{\varepsilon_0 w}{d}\big[\ell + (\epsilon - 1)x\big], \qquad \frac{\mathrm{d}C}{\mathrm{d}x} = \frac{\varepsilon_0 w(\epsilon - 1)}{d}.$$

Constant Q. $U = Q^2/2C$. Differentiating, $\mathrm{d}U/\,\mathrm{d}x = -\dfrac{Q^2}{2C^2}\dfrac{\mathrm{d}C}{\mathrm{d}x}$, so the force in the direction of increasing x is

$$F = -\left(\frac{\mathrm{d}U}{\mathrm{d}x}\right)_Q = \frac{Q^2}{2C^2}\frac{\mathrm{d}C}{\mathrm{d}x} = \tfrac{1}{2}V^2\frac{\mathrm{d}C}{\mathrm{d}x} \qquad (V = Q/C).$$

Constant V. By Problem 11(b), including the battery's work, $F = +\left(\dfrac{\mathrm{d}}{\mathrm{d}x}\tfrac{1}{2}CV^2\right)_V = \tfrac{1}{2}V^2\dfrac{\mathrm{d}C}{\mathrm{d}x}$.

Result. Both give

$$F = \tfrac{1}{2}V^2 \cdot \frac{\varepsilon_0 w(\epsilon - 1)}{d} = \frac{\varepsilon_0 w(\epsilon - 1)V^2}{2d},$$

positive, i.e. pulling the slab *in*. The force does not depend on x. Physically it comes from the fringing field at the plate edges, which polarises the slab and pulls on its bound charges; the energy method gets the answer without needing to know the fringing field in detail.

11.2 Charged soap bubble

Show that the outward pressure on a charged soap bubble is $\sigma^2/(2\varepsilon_0)$ and find the charge needed to change its radius by 1% given surface tension γ.

Pressure. The soap film conducts, so $E = 0$ inside and $E = \sigma/\varepsilon_0$ just outside. The charge feels the average field $\sigma/2\varepsilon_0$ (it does not act on itself; Problem 11), giving an outward pressure $\sigma \cdot \sigma/2\varepsilon_0 = \sigma^2/2\varepsilon_0$, exactly as for any conductor.

Balance. Without charge, $p_{\text{in}} - p_{\text{out}} = 4\gamma/R$. The electrostatic pressure pushes outwards, so it supplies part of the outward force and less gas pressure is needed:

$$p_{\text{in}} - p_{\text{out}} = \frac{4\gamma}{R} - \frac{\sigma^2}{2\varepsilon_0}.$$

Enclosed gas. For a closed bubble the gas obeys $p_{\text{in}} R^3 \approx \text{const}$ (isothermal, volume $\propto R^3$). Differentiating, $\Delta p_{\text{in}}/p_{\text{in}} = -3\Delta R/R$, so a 1% increase in R lowers p_{in} by about 3% of an atmosphere, 3×10^3 Pa. The surface-tension term $4\gamma/R$ is tiny for a soap bubble (10 Pa or so) and changes negligibly. So we need $\sigma^2/2\varepsilon_0 \approx 3 \times 10^3$ Pa:

$$E = \frac{\sigma}{\varepsilon_0} = \sqrt{\frac{2 \times 3 \times 10^3}{\varepsilon_0}} = \sqrt{\frac{6 \times 10^3}{8.85 \times 10^{-12}}}\ \mathrm{V\,m^{-1}} = 2.6 \times 10^7\ \mathrm{V\,m^{-1}}.$$

This is nearly ten times the breakdown field of air, so a 1% expansion cannot be reached: the air sparks first. At the breakdown limit ($\sigma^2/2\varepsilon_0 \approx 40$ Pa) the radius changes by only $40/(3 \times 10^5) \approx 10^{-4}$. The lesson: electrostatic pressure is tiny compared with atmospheric pressure. (Only a bubble whose gas can escape, e.g. one still on its pipe, responds more strongly.)

11.3 The Maxwell stress tensor

Write the force on a conductor as the integral of the Maxwell stress tensor and show it reproduces $\sigma^2/(2\varepsilon_0)$ normal to the surface.

Stress tensor. The total electric force on the charges inside a closed surface S can be written as a surface integral of the fields on S alone:

$$F_i = \oint_S T_{ij} n_j \,\mathrm{d}A, \qquad T_{ij} = \varepsilon_0 \left(E_i E_j - \tfrac{1}{2}\delta_{ij} E^2 \right),$$

with summation over j, and $\hat{\mathbf{n}}$ the outward normal.

Apply it. Wrap S tightly round the conductor, just outside its surface. There $\mathbf{E} = (\sigma/\varepsilon_0)\hat{\mathbf{n}}$, so $\mathbf{E} \cdot \hat{\mathbf{n}} = E$ and $E_i = E n_i$:

$$T_{ij} n_j = \varepsilon_0 \left(E_i (\mathbf{E} \cdot \hat{\mathbf{n}}) - \tfrac{1}{2} E^2 n_i \right) = \varepsilon_0 \left(E^2 n_i - \tfrac{1}{2} E^2 n_i \right) = \tfrac{1}{2}\varepsilon_0 E^2 n_i = \frac{\sigma^2}{2\varepsilon_0}\, n_i.$$

Each element of surface is pulled outward along its normal with stress $\sigma^2/2\varepsilon_0$, as found from energy. The stress tensor makes Faraday's picture quantitative: field lines carry a *tension* $\frac{1}{2}\varepsilon_0 E^2$ along their length and exert a *pressure* $\frac{1}{2}\varepsilon_0 E^2$ sideways.

Problem 12 — Electrostatic balance

What you need to know for these extensions

- Problem 12: the net closing force $G(x)$, the equilibrium condition $k(a-b) = \varepsilon_0 A V^2/(2b^2)$, and stability from the sign of $G'(b)$.
- Problem 11(a): at constant charge the force between the plates is $Q^2/(2\varepsilon_0 A)$, independent of separation.
- **Simple harmonic motion:** a mass m with restoring force $-k_{\text{eff}}\,\delta$ oscillates at $\omega = \sqrt{k_{\text{eff}}/m}$.

12.1 Constant charge

Repeat at constant charge*: show the system is then stable for all b. Why does the electrical boundary condition matter so much?*

At constant Q the force is $F = Q^2/2\varepsilon_0 A$ (Problem 11(a)), *independent of the gap*. The net closing force is

$$G(x) = \frac{Q^2}{2\varepsilon_0 A} - k(a-x), \qquad \frac{\mathrm{d}G}{\mathrm{d}x} = +k > 0 \text{ for every } x,$$

so every equilibrium is stable, right down to contact.

Why it matters. At constant V, closing the gap raises C, so the battery pushes in more charge, which raises the force (as $1/x^2$): positive feedback. At constant Q there is no feedback, because the field between the plates is fixed by $\sigma = Q/A$. MEMS designers use charge drive (or series capacitors) to extend the stable travel beyond one third of the gap.

12.2 Small oscillations

Find the frequency of small oscillations about the stable equilibrium as a function of b/a, and show it goes to zero at pull-in.

Effective spring constant. Displace the disc to $x = b + \delta$. The net force tending to close the gap is $G(b+\delta) \approx G'(b)\,\delta$, so the restoring force (tending to *return* it) is $-G'(b)\,\delta$, a spring of stiffness

$$k_{\text{eff}} = G'(b) = k - \frac{\varepsilon_0 A V^2}{b^3} = k - \frac{2kb^2(a-b)}{b^3} = k\left(1 - \frac{2(a-b)}{b}\right) = k\,\frac{3b-2a}{b},$$

using the equilibrium condition $\varepsilon_0 A V^2 = 2kb^2(a-b)$.

Frequency. For an upper disc of mass m, with $\omega_0 = \sqrt{k/m}$ the frequency with no voltage:

$$\omega = \sqrt{\frac{k_{\text{eff}}}{m}} = \omega_0\sqrt{\frac{3b-2a}{b}} = \omega_0\sqrt{3 - \frac{2a}{b}}.$$

At $V = 0$ ($b = a$), $\omega = \omega_0$; as $b \to \frac{2}{3}a$, $3 - 2a/b \to 0$ and $\omega \to 0$. This "electrostatic spring softening" is used to tune MEMS resonators, and the vanishing frequency signals the approach to pull-in.

12.3 A MEMS switch

Estimate V_c for a MEMS switch with a $2\,\mu\text{m}$ *gap,* $100\,\mu\text{m}$ *square plate and* $k = 1\,\text{N}\,\text{m}^{-1}$.

From Problem 12, $V_c^2 = 8ka^3/(27\varepsilon_0 A)$, with $a = 2\times10^{-6}\,\mathrm{m}$ and $A = (10^{-4}\,\mathrm{m})^2 = 10^{-8}\,\mathrm{m}^2$:

$$V_c = \sqrt{\frac{8(1)(2\times10^{-6})^3}{27(8.85\times10^{-12})(10^{-8})}}\,\mathrm{V} = \sqrt{\frac{6.4\times10^{-17}}{2.39\times10^{-18}}}\,\mathrm{V} = \sqrt{26.8}\,\mathrm{V} = 5.2\,\mathrm{V}.$$

A few volts — which is why electrostatic actuation is practical at micrometre scales but needs kilovolts for millimetre gaps ($V_c \propto a^{3/2}$).

Problem 13 — Coaxial cable

What you need to know for these extensions

- Problem 13: $D = \lambda/(2\pi s)$ in both layers, $E = D/(\epsilon_i\varepsilon_0)$, and $V = (\lambda/2\pi\varepsilon_0)\,S$ with $S = \ln(c/a)/\epsilon_1 + \ln(b/c)/\epsilon_2$.
- **Boundary conditions** at a dielectric interface: tangential **E** continuous, normal **D** continuous (Problem 14).
- **Polarisation and bound charge:** $\mathbf{P} = \mathbf{D} - \varepsilon_0\mathbf{E} = \varepsilon_0(\epsilon - 1)\mathbf{E}$; a bound surface charge $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$ appears on a dielectric surface with outward normal $\hat{\mathbf{n}}$, and a bound volume charge $\rho_b = -\nabla\cdot\mathbf{P}$.

13.1 Field grading

Find the maximum field in each layer per unit voltage, and choose c to make the two maxima equal.

Step 1. In layer i, $E = \lambda/(2\pi\epsilon_i\varepsilon_0 s)$, which is largest at the layer's inner radius ($s = a$ for layer 1, $s = c$ for layer 2). From $V = (\lambda/2\pi\varepsilon_0)S$, $\lambda/(2\pi\varepsilon_0) = V/S$, so

$$\frac{E_{1,\max}}{V} = \frac{1}{\epsilon_1 a S}, \qquad \frac{E_{2,\max}}{V} = \frac{1}{\epsilon_2 c S}.$$

Step 2: original design ($c = 3\,\mathrm{mm}$, $S = 0.502$):

$$\frac{E_{1,\max}}{V} = \frac{1}{5(1\times10^{-3})(0.502)} = 398\,\mathrm{m}^{-1}, \qquad \frac{E_{2,\max}}{V} = \frac{1}{3(3\times10^{-3})(0.502)} = 221\,\mathrm{m}^{-1}.$$

Layer 1 is the weak point.

Step 3: equal maxima. $\epsilon_1 a S = \epsilon_2 c S$ gives $c = \epsilon_1 a/\epsilon_2 = 5\times1\,\mathrm{mm}/3 = 1.67\,\mathrm{mm}$. Then $S = \ln(1.67)/5 + \ln(7/1.67)/3 = 0.102 + 0.478 = 0.581$ and both maxima are $1/(5\times1\times10^{-3}\times0.581) = 345\,\mathrm{m}^{-1}$. A single dielectric gives $1/(a\ln(b/a)) = 514\,\mathrm{m}^{-1}$ whatever its permittivity, so graded insulation lowers the peak field by a third.

13.2 Dielectrics as sectors

Suppose instead the dielectrics were arranged as sectors (dielectric 1 filling a 60° wedge from a to b). Show the capacitors are then in parallel and find C′.

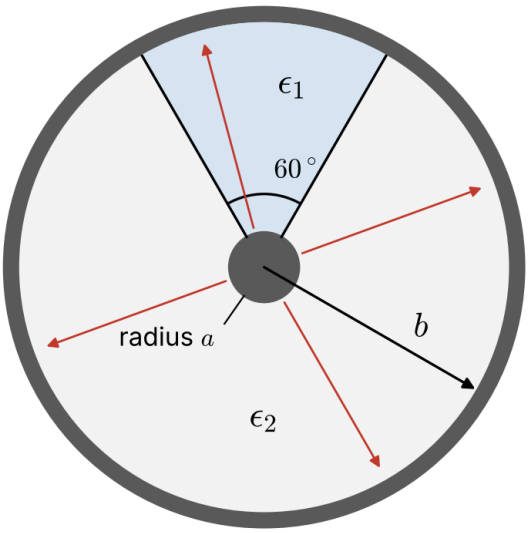


Figure 11: Dielectrics arranged as sectors: dielectric 1 fills a 60° wedge, dielectric 2 the rest. The radial **E** is the same in both, so the two parts act as capacitors in parallel.

Step 1: E **is the same in both sectors.** Each sector spans from the inner to the outer conductor, so each sees the full voltage V. At the radial interfaces between sectors, the radial **E** is *tangential* to the interface and must be continuous. So E is the same in both, $E = V/(s\ln(b/a))$ (the single-dielectric form), while $D = \epsilon_i\varepsilon_0 E$ differs.

Step 2: add the charges. The free charge per unit length on the inner conductor is D integrated round it: a fraction $\frac{1}{6}$ of the circumference with ϵ_1, $\frac{5}{6}$ with ϵ_2:

$$\lambda = 2\pi a\left[\tfrac{1}{6}\epsilon_1 + \tfrac{5}{6}\epsilon_2\right]\varepsilon_0\frac{V}{a\ln(b/a)} \quad\Longrightarrow\quad C' = \frac{\lambda}{V} = \frac{2\pi\varepsilon_0}{\ln(b/a)}\left[\tfrac{1}{6}\epsilon_1 + \tfrac{5}{6}\epsilon_2\right].$$

The capacitances of the two sectors add: they are in parallel. Numerically, $\frac{1}{6}(5) + \frac{5}{6}(3) = 3.33$ and

$$C' = \frac{5.56\times10^{-11}\,\mathrm{F\,m^{-1}}}{1.946}\times 3.33 = 95\,\mathrm{pF\,m^{-1}}.$$

This is less than the 111 $\mathrm{pF\,m^{-1}}$ of the layered cable: there the high-permittivity material sits where the field is strongest, so it counts for more.

13.3 Bound charges

Compute the bound surface charge densities on all three surfaces and check that the total bound charge is zero.

Rule. $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$, with $\hat{\mathbf{n}}$ the outward normal of the dielectric, and $\mathbf{P} = \varepsilon_0(\epsilon-1)\mathbf{E} = \dfrac{\epsilon-1}{\epsilon}\mathbf{D} = \dfrac{\epsilon-1}{\epsilon}\dfrac{\lambda}{2\pi s}\hat{\mathbf{s}}$. There is no volume bound charge: $\nabla\cdot\mathbf{P} \propto \nabla\cdot(\hat{\mathbf{s}}/s) = \frac{1}{s}\frac{\mathrm{d}}{\mathrm{d}s}(s\cdot\frac{1}{s}) = 0$. The bound charge per unit length on a surface of radius s is $\sigma_b\times 2\pi s$. On the inner face of a layer the outward normal is $-\hat{\mathbf{s}}$; on the outer face it is $+\hat{\mathbf{s}}$.

Surface	σ_b	Bound charge per unit length
$s = a$ (dielectric 1, inner face)	$-\dfrac{\epsilon_1 - 1}{\epsilon_1}\dfrac{\lambda}{2\pi a}$	$-\frac{4}{5}\lambda = -0.80\lambda$
$s = c$ (dielectric 1 side)	$+\dfrac{\epsilon_1 - 1}{\epsilon_1}\dfrac{\lambda}{2\pi c}$	$+0.80\lambda$
$s = c$ (dielectric 2 side)	$-\dfrac{\epsilon_2 - 1}{\epsilon_2}\dfrac{\lambda}{2\pi c}$	$-\frac{2}{3}\lambda = -0.67\lambda$
$s = b$ (dielectric 2, outer face)	$+\dfrac{\epsilon_2 - 1}{\epsilon_2}\dfrac{\lambda}{2\pi b}$	$+0.67\lambda$

The total is zero, as it must be for a neutral dielectric. The net bound charge at the interface $s = c$ is $(0.80 - 0.67)\lambda = +0.13\lambda$, as found in Problem 13. Check: inside dielectric 1, Gauss's law with free plus bound charge enclosed, $\lambda - 0.8\lambda = \lambda/5$, gives $E = \lambda/(2\pi\varepsilon_0 \cdot 5s)$ — the field found with $\mathbf{D}$.

Problem 14 — Tilted cavity

What you need to know for these extensions

- Problem 14: at each interface, tangential $\mathbf{E}$ and normal $\mathbf{D}$ are continuous; components relative to a tilted surface are found by dot products with its unit normal and tangent.
- **Additional knowledge for 14.2:** the field inside a long cylindrical cavity in a dielectric, with the applied field across the cylinder, is $2\epsilon/(\epsilon + 1)$ times the field in the dielectric. (This is the two-dimensional analogue of the sphere result of Problem 7 and extension 7.3, obtained by the same method with cylindrical harmonics.)
- Bound surface charge $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ (Problem 13 background).

14.1 Dependence on the tilt

Find E_1/E_0 as a function of the tilt angle β of the cavity (keeping $\mathbf{E}_0$ along the cavity normal), and plot it from 0 to 90°.

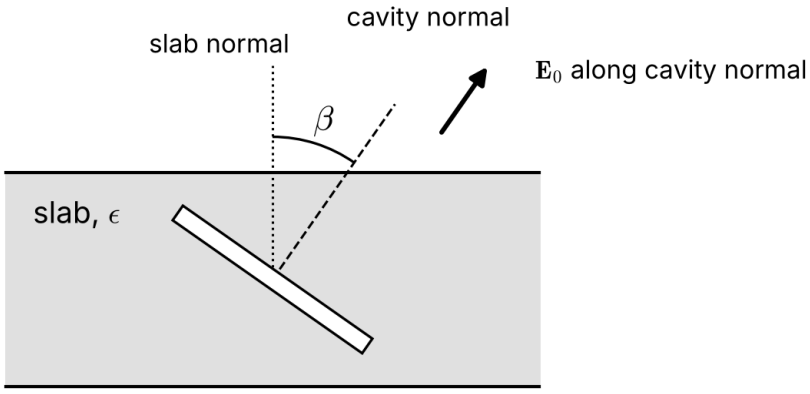


Figure 12: The cavity tilted by β, the angle between the slab normal and the cavity normal. $\mathbf{E}_0$ is kept along the cavity normal.

Step 1: field in the slab. With the slab normal along $\hat{\mathbf{z}}$, the cavity normal and tangent are

$\hat{\mathbf{n}}_c = (\sin\beta, 0, \cos\beta)$ and $\hat{\mathbf{t}}_c = (\cos\beta, 0, -\sin\beta)$, and $\mathbf{E}_0 = E_0\hat{\mathbf{n}}_c$. At the slab face E_x is continuous and E_z is divided by ϵ:

$$\mathbf{E}_d = E_0\left(\sin\beta,\ 0,\ \frac{\cos\beta}{\epsilon}\right).$$

Step 2: components relative to the cavity.

$$E_{d,n} = \mathbf{E}_d\cdot\hat{\mathbf{n}}_c = E_0\left(\sin^2\beta + \frac{\cos^2\beta}{\epsilon}\right), \qquad E_{d,t} = \mathbf{E}_d\cdot\hat{\mathbf{t}}_c = E_0\left(\sin\beta\cos\beta - \frac{\sin\beta\cos\beta}{\epsilon}\right)$$
$$= E_0\sin\beta\cos\beta\left(1-\frac{1}{\epsilon}\right).$$

Step 3: field in the cavity. The normal component is multiplied by ϵ on entering the vacuum; the tangential one is unchanged:

$$\frac{E_1}{E_0} = \sqrt{\left(\epsilon\sin^2\beta + \cos^2\beta\right)^2 + \sin^2\beta\cos^2\beta\left(1-\frac{1}{\epsilon}\right)^2}.$$

Checks. At $\beta = 0$ (cavity parallel to the slab): $E_1/E_0 = 1$. At $\beta = 90°$: $E_1/E_0 = \epsilon = 2$. At $\beta = 45°$, $\epsilon = 2$: $\sqrt{(1+\frac{1}{2})^2 + \frac{1}{4}\cdot\frac{1}{4}} = \sqrt{37}/4 = 1.52$, as in Problem 14. The ratio rises smoothly between these (Figure 13).

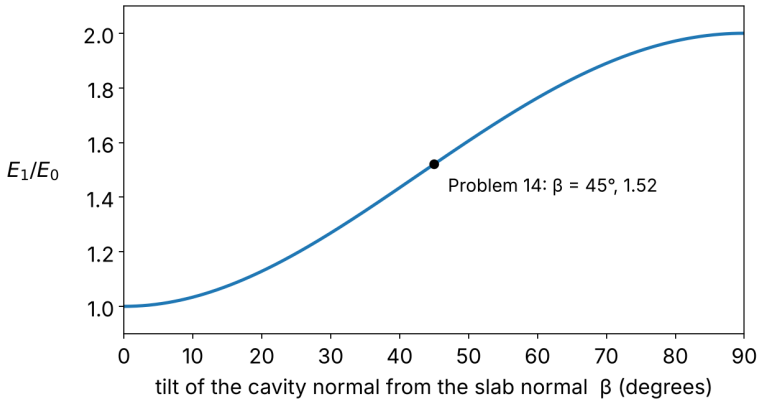


Figure 13: Field in the cavity as the cavity is tilted, for $\epsilon = 2$.

14.2 Needle-shaped cavity

Replace the thin disc by a needle-shaped cavity along the same direction, and find E_1. Which shape gives the larger field?

The needle lies along $\hat{\mathbf{n}}_c$. Split the slab field into components along and across the needle (from Problem 14, Step 2: $E_{d,n} = \frac{3}{4}E_0$ along, $E_{d,t} = \frac{1}{4}E_0$ across).

Step 1: along the needle. The long sides of the needle are parallel to $\hat{\mathbf{n}}_c$, so this component is tangential to them and continuous: $E_{1,\parallel} = E_{d,n} = \frac{3}{4}E_0$.

Step 2: across the needle. A transverse field in a thin cylindrical cavity is enhanced by $2\epsilon/(\epsilon+1)$ (background): $E_{1,\perp} = \dfrac{2\epsilon}{\epsilon+1}E_{d,t} = \dfrac{4}{3}\cdot\dfrac{1}{4}E_0 = \dfrac{1}{3}E_0$.

Result. $E_1 = E_0\sqrt{(3/4)^2 + (1/3)^2} = 0.82E_0$, against $1.52E_0$ for the disc. A flat cavity perpendicular to $\mathbf{D}$ is the dangerous shape: it carries the full D into vacuum. A needle along the field sees only the (reduced) field of the dielectric. This is why delamination voids in insulation are so harmful.

14.3 Bound surface charges

Find the bound surface charge densities on the slab faces and on the cavity faces.

Use $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$ with $\mathbf{P} = \varepsilon_0(\epsilon-1)\mathbf{E}_d = \varepsilon_0\mathbf{E}_d$ for $\epsilon = 2$, and $\hat{\mathbf{n}}$ pointing out of the dielectric.

- **Slab faces** ($\hat{\mathbf{n}} = \pm\hat{\mathbf{z}}$): $P_z = \varepsilon_0 E_{d,z} = \varepsilon_0 E_0/(2\sqrt{2})$, so $\sigma_b = \pm 0.35\,\varepsilon_0 E_0$ (positive on the face that $\mathbf{E}_0$ leaves through). Check: the normal field jumps from $E_0/\sqrt{2}$ outside to $E_0/(2\sqrt{2})$ inside, a jump of $E_0/(2\sqrt{2}) = \sigma_b/\varepsilon_0$. ✓
- **Cavity faces** ($\hat{\mathbf{n}} = \pm\hat{\mathbf{n}}_c$, pointing out of the dielectric into the cavity): $\sigma_b = \pm P_n = \pm\varepsilon_0 E_{d,n} = \pm 0.75\,\varepsilon_0 E_0$. Check: the normal field jumps from $0.75E_0$ in the dielectric to $1.5E_0$ in the cavity, a jump of $0.75E_0 = |\sigma_b|/\varepsilon_0$. ✓

Problem 15 — Dielectric relaxation

What you need to know for these extensions

- Problem 15: continuity equation, Ohm's law and Gauss's law combined into $\partial\rho/\partial t = -\rho/\tau$, $\tau = \epsilon\varepsilon_0/\sigma$.
- **Diffusion:** a concentration gradient drives a flux $-D_c\nabla\rho$ (Fick's law).
- **Additional knowledge for 15.1:** the Einstein relation $\sigma = ne^2 D_c/(k_B T)$ between conductivity and diffusion coefficient (statistical physics).
- **For 15.2:** the plasma frequency $\omega_p = \sqrt{ne^2/\varepsilon_0 m_e}$, derived in Sheet 2, Problem 35.

15.1 Diffusion and the Debye length

Include diffusion of charge carriers, $\mathbf{J} = \sigma\mathbf{E} - D_c\nabla\rho$ with D_c the diffusion coefficient. Show that a static charge perturbation is screened over the Debye length $\lambda_D = \sqrt{\epsilon\varepsilon_0 D_c/\sigma}$.

Step 1: continuity with the new current. Take the divergence of $\mathbf{J}$, using $\nabla\cdot\mathbf{E} = \rho/\epsilon\varepsilon_0$ and $\nabla\cdot\nabla\rho = \nabla^2\rho$:

$$\frac{\partial\rho}{\partial t} = -\nabla\cdot\mathbf{J} = -\sigma\nabla\cdot\mathbf{E} + D_c\nabla^2\rho = -\frac{\rho}{\tau} + D_c\nabla^2\rho, \qquad \tau = \frac{\epsilon\varepsilon_0}{\sigma}.$$

Step 2: static state. Setting $\partial\rho/\partial t = 0$:

$$\nabla^2\rho = \frac{\rho}{D_c\tau} = \frac{\rho}{\lambda_D^2}, \qquad \lambda_D = \sqrt{D_c\tau} = \sqrt{\frac{\epsilon\varepsilon_0 D_c}{\sigma}}.$$

In one dimension, $\mathrm{d}^2\rho/\,\mathrm{d}x^2 = \rho/\lambda_D^2$ has the decaying solution $\rho \propto e^{-x/\lambda_D}$; in three dimensions, around a point disturbance, $\rho \propto e^{-r/\lambda_D}/r$. Either way the charge is screened over the distance λ_D.

Step 3: the familiar form. With the Einstein relation $\sigma = ne^2 D_c/k_B T$, $D_c/\sigma = k_B T/(ne^2)$ and

$$\lambda_D = \sqrt{\frac{\epsilon\varepsilon_0 k_B T}{ne^2}},$$

the Debye length of plasma and semiconductor physics. For silicon ($\epsilon = 11.7$) with $n = 10^{22}\,\mathrm{m}^{-3}$ at room temperature ($k_B T/e = 0.0259\,\mathrm{V}$),

$$\lambda_D = \sqrt{\frac{(11.7)(8.85\times10^{-12})(0.0259)}{(10^{22})(1.602\times10^{-19})}}\ \mathrm{m} \approx 41\,\mathrm{nm}.$$

15.2 How fast does a metal neutralise?

A metal sphere is suddenly given a charge at its centre. Estimate, using the corrected picture for metals, how long the charge takes to reach the surface.

- **Locally**, the excess charge is neutralised by a tiny displacement of the surrounding electrons, which overshoot and oscillate at the plasma frequency $\omega_p = \sqrt{ne^2/\varepsilon_0 m_e}$. For copper ($n = 8.5 \times 10^{28}\,\mathrm{m}^{-3}$),

$$\omega_p = \sqrt{\frac{(8.5\times10^{28})(1.602\times10^{-19})^2}{(8.85\times10^{-12})(9.11\times10^{-31})}}\,\mathrm{s}^{-1} = 1.6\times10^{16}\,\mathrm{s}^{-1},$$

  so neutralisation takes $\sim 1/\omega_p \sim 10^{-16}\,\mathrm{s}$. The oscillation is damped on the collision time, $\sim 10^{-14}\,\mathrm{s}$.

- **Globally**, no individual electron travels from the centre to the surface. The charge "reaches" the surface as a wave of tiny displacements, and the final surface distribution is established in about the light-travel time across the sphere, $R/c \approx 30\,\mathrm{ps}$ for $R = 1\,\mathrm{cm}$.

The naive $\varepsilon_0/\sigma \approx 10^{-19}\,\mathrm{s}$ is unphysical: it is shorter than the time electrons take to respond at all.

15.3 Seawater: conductor or dielectric?

At what frequency does seawater change from behaving as a conductor to behaving as a dielectric? Relate this to why submarines communicate at very low frequencies.

In the Ampère–Maxwell law, $\nabla\times\mathbf{H} = \sigma\mathbf{E} + \epsilon\varepsilon_0\partial\mathbf{E}/\partial t$; for $e^{-i\omega t}$ fields the conduction current σE equals the displacement current $\epsilon\varepsilon_0\omega E$ in magnitude when

$$f = \frac{\sigma}{2\pi\epsilon\varepsilon_0} = \frac{4}{2\pi(80)(8.85\times10^{-12})}\,\mathrm{Hz} = 0.9\,\mathrm{GHz}.$$

Below about 1 GHz seawater is a conductor, with skin depth $\delta = \sqrt{2/\mu_0\sigma\omega}$ growing as $f^{-1/2}$. Above it, seawater is a lossy dielectric whose attenuation saturates at $\alpha = \frac{\sigma}{2}\sqrt{\mu_0/\epsilon\varepsilon_0} = \frac{\sigma Z_0}{2\sqrt{\epsilon}} \approx 84\,\mathrm{m}^{-1}$ (Sheet 2, Problem 38(b)), a $1/e$ depth of about 1 cm. Either way, penetration improves only by going to *lower* frequency, which is why submarines use VLF and ELF (Problem 37).

Problem 16 — Straight wire

What you need to know for these extensions

- The Biot–Savart law and the geometry of Problem 16; the angle integrals of extension 1.1.
- The force per unit length between parallel currents, $\mu_0 I_1 I_2/(2\pi d)$, and between parallel line charges, $\lambda_1\lambda_2/(2\pi\varepsilon_0 d)$ (from the field of a line charge, Problem 1).
- $\mu_0\varepsilon_0 = 1/c^2$.

16.1 Finite wire segment

Find B at distance r from a finite *wire segment, and compare with Problem 1(a).*

In Problem 16, an element $\mathrm{d}z$ of wire at displacement $\mathbf{R}$ from the field point contributes $|\,\mathrm{d}\mathbf{B}| = \mu_0 I\,\mathrm{d}z\sin\phi/(4\pi R^2)$ with $\sin\phi = r/R$, i.e. $\mu_0 I\,r\,\mathrm{d}z/(4\pi R^3)$, all in the same direction. This is exactly the integrand of $E_\perp$ in extension 1.1 ($\lambda z\,\mathrm{d}x'/4\pi\varepsilon_0 R^3$) with $\lambda/\varepsilon_0 \to \mu_0 I$ and $z \to r$. So, with α_1, α_2 the angles subtended by the ends, measured from the perpendicular,

$$B = \frac{\mu_0 I}{4\pi r}\,(\sin\alpha_2 - \sin\alpha_1).$$

For a symmetric segment of length $2L$, $\sin\alpha_2 = -\sin\alpha_1 = L/\sqrt{L^2+r^2}$ and $B = \dfrac{\mu_0 I}{2\pi r}\dfrac{L}{\sqrt{L^2+r^2}}$, the exact analogue of Problem 1(a); for $L \to \infty$ it gives $\mu_0 I/2\pi r$. A segment alone cannot carry a steady current, so this is the contribution of one straight piece of a closed circuit (useful for square loops, for example).

16.2 Busbar forces

> *Two parallel wires carry* 1 kA *in opposite directions,* 1 cm *apart, as in a power busbar. Find the force per metre, and comment on short-circuit forces.*

$$\frac{F}{l} = \frac{\mu_0 I^2}{2\pi d} = \frac{(2\times10^{-7}\,\mathrm{N\,A^{-2}})(10^3\,\mathrm{A})^2}{0.01\,\mathrm{m}} = 20\,\mathrm{N\,m^{-1}},$$

repulsive (antiparallel currents). The force grows as I^2: a 50 kA short-circuit fault gives $50^2 = 2500$ times more, $50\,\mathrm{kN\,m^{-1}}$. Busbars must be braced for fault currents, not for normal operation.

16.3 Two electron beams

> *Explain why the force between two parallel beams of electrons moving at speed v is reduced by the factor $(1 - v^2/c^2)$ relative to pure electrostatic repulsion.*

Model each beam as a line charge λ carrying current $I = \lambda v$, separated by d. The two forces per unit length are

$$\frac{F_E}{l} = \lambda\cdot\frac{\lambda}{2\pi\varepsilon_0 d} = \frac{\lambda^2}{2\pi\varepsilon_0 d}\ \text{(repulsive)}, \qquad \frac{F_B}{l} = \frac{\mu_0(\lambda v)^2}{2\pi d} = \frac{\lambda^2}{2\pi\varepsilon_0 d}\cdot\mu_0\varepsilon_0 v^2$$

$$= \frac{\lambda^2}{2\pi\varepsilon_0 d}\,\frac{v^2}{c^2}\ \text{(attractive)},$$

the magnetic force being attractive because the currents are parallel. So the net repulsion is $F/l = \dfrac{\lambda^2}{2\pi\varepsilon_0 d}\left(1 - \dfrac{v^2}{c^2}\right)$. The magnetic force is the relativistic correction to the electric one: in the beams' rest frame there is only electrostatic repulsion, and transforming the force back to the lab gives the same factor. Ultra-relativistic beams barely repel each other, which is what makes high-current accelerator beams possible.

Problem 17 — Loops and solenoids

What you need to know for these extensions

- The on-axis loop field $B = \mu_0 I a^2/[2(a^2+u^2)^{3/2}]$ and the slice-by-slice construction of a solenoid, Problem 17.

- The integral $\int \mathrm{d}u/(a^2+u^2)^{3/2} = u/(a^2\sqrt{a^2+u^2})$ (Problem 1).
- **Even functions:** if $B(-x) = B(x)$, all odd derivatives of B vanish at $x = 0$.

17.1 Finite solenoid

Find the field on the axis of a finite solenoid of length ℓ in terms of the angles subtended by its ends, and plot B along the axis.

Step 1: sum the loops. Measure positions from the solenoid's centre; the windings occupy $-\ell/2 \le x' \le \ell/2$ and the field point is at x. The slice at x' carries current $nI\,\mathrm{d}x'$ at axial distance $u = x' - x$, so

$$B(x) = \frac{\mu_0 n I a^2}{2} \int_{-\ell/2-x}^{\ell/2-x} \frac{\mathrm{d}u}{(a^2+u^2)^{3/2}} = \frac{\mu_0 n I}{2} \left[\frac{u}{\sqrt{a^2+u^2}} \right]_{-\ell/2-x}^{\ell/2-x},$$

$$B(x) = \frac{\mu_0 n I}{2} \left[\frac{\ell/2-x}{\sqrt{(\ell/2-x)^2+a^2}} + \frac{\ell/2+x}{\sqrt{(\ell/2+x)^2+a^2}} \right] = \frac{\mu_0 n I}{2} (\cos\theta_1 + \cos\theta_2),$$

where θ_1, θ_2 are the angles between the axis and the lines from the field point to the rims of the two end faces (each fraction is adjacent over hypotenuse in the right-angled triangle formed by the axis, the end face's radius and that line).

Step 2: checks. Deep inside a long solenoid both angles $\to 0$ and $B \to \mu_0 n I$. At one end of a long solenoid one angle is 0 and the other 90°, giving $\frac{1}{2}\mu_0 n I$, as in Problem 17(b). Figure 14 shows that a solenoid needs $\ell \gtrsim 10a$ before its central field approaches the ideal value.

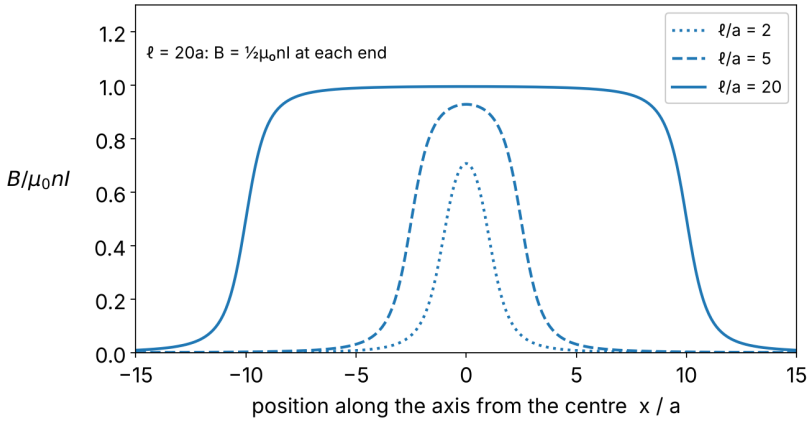


Figure 14: Field on the axis of finite solenoids of radius a.

17.2 Where is the force largest?

Where on the axis, outside the solenoid, is the force on the dipole largest?

From Problem 17(b), $|F| = \mu_0 n m I a^2/[2(x_0^2+a^2)^{3/2}]$, which decreases steadily as x_0 increases. So outside the solenoid the force is largest right at the end, $x_0 = 0$, where $F = \mu_0 n m I a^2/(2a^3) = \mu_0 n m I/2a$. Inside a semi-infinite solenoid, at depth x_0, the same calculation gives the same formula (the field is $\frac{1}{2}\mu_0 n I(1 + x_0/\sqrt{x_0^2+a^2})$, whose slope has the same magnitude), so the force peaks exactly at the end face and falls off on both sides. Deep inside, where B is uniform, there is no force at all.

17.3 Helmholtz coils

Show that two coaxial loops separated by their radius (a Helmholtz pair) produce a field with zero first, second and third derivatives at the midpoint.

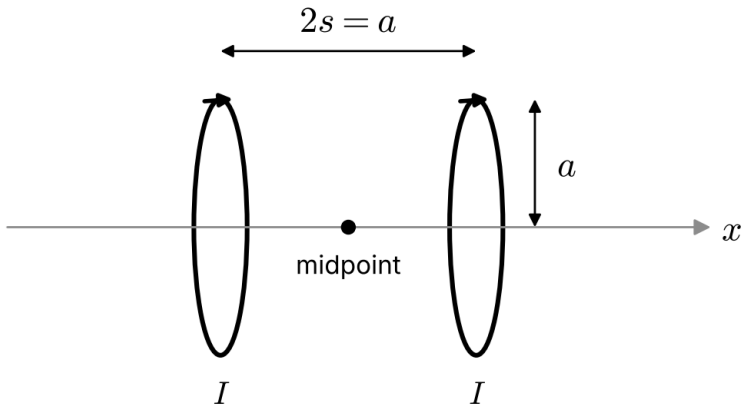


Figure 15: Helmholtz coils: two coaxial loops of radius a, separated by $2s = a$, carrying equal currents in the same sense.

Set-up. Loops of radius a at $x = \pm s$, each carrying I in the same sense. Write the single-loop field as $\frac{1}{2}\mu_0 I a^2 f(u)$ with $f(u) = (a^2 + u^2)^{-3/2}$, u the distance from that loop. Then

$$B(x) = \tfrac{1}{2}\mu_0 I a^2 \big[f(x-s) + f(x+s)\big].$$

Odd derivatives. f is even, so $B(-x) = \frac{1}{2}\mu_0 I a^2[f(-x-s) + f(-x+s)] = \frac{1}{2}\mu_0 I a^2[f(x+s) + f(x-s)] = B(x)$. An even function has zero first and third derivatives at $x = 0$, for any s.

Second derivative. Differentiate f twice (product and chain rules):

$$f'(u) = -3u(a^2 + u^2)^{-5/2},$$

$$f''(u) = -3(a^2 + u^2)^{-5/2} + 15u^2(a^2 + u^2)^{-7/2} = 3(a^2 + u^2)^{-7/2}(4u^2 - a^2).$$

At the midpoint $B''(0) = \frac{1}{2}\mu_0 I a^2[f''(-s) + f''(s)] = \mu_0 I a^2 f''(s)$, which vanishes when $4s^2 = a^2$, i.e. $s = a/2$: the loops are separated by $2s = a$, their radius. ✓

The first non-zero derivative is then the fourth, so the field is uniform to fourth order. Its centre value is $B = \mu_0 I a^2 f(a/2) = \mu_0 I a^2 (5a^2/4)^{-3/2} = (4/5)^{3/2}\mu_0 I/a = 0.716\,\mu_0 I/a$. Helmholtz coils are the standard way to make a uniform field in a large accessible volume.

Problem 18 — Wire with a cavity

What you need to know for these extensions

- Problem 18: superposition of $+J$ and $-J$ wires. Problem 18(a)'s results written as vectors are $\mathbf{B}_{\text{in}} = \frac{1}{2}\mu_0 \mathbf{J} \times \mathbf{s}$, $\mathbf{B}_{\text{out}} = \frac{1}{2}\mu_0 R^2 \mathbf{J} \times \mathbf{s}/s^2$ for a wire of radius R, where $\mathbf{s}$ is measured from that wire's axis.
- Cross products: $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$, $\hat{\mathbf{z}} \times \hat{\mathbf{y}} = -\hat{\mathbf{x}}$.
- **For 18.3:** a surface current density K (current per unit width) as the limit of a thin layer of current density J and thickness t: $K = Jt$.

18.1 In terms of the total current

Express the answer in terms of the total current I and find B_A and B_B numerically for $I = 10\,\text{A}$, $a = 1\,\text{mm}$.

The conducting area is $\pi a^2 - \pi(a/2)^2 = \frac{3}{4}\pi a^2$, so $J = I/(\frac{3}{4}\pi a^2) = 4I/(3\pi a^2)$. Substituting into $B_A = \mu_0 J a/4$ and $B_B = 5\mu_0 J a/12$:

$$B_A = \frac{\mu_0 J a}{4} = \frac{\mu_0 I}{3\pi a} = \frac{(4\pi \times 10^{-7})(10)}{3\pi(10^{-3})}\,\text{T} = 1.33\,\text{mT}, \qquad B_B = \frac{5\mu_0 J a}{12} = \frac{5\mu_0 I}{9\pi a} = 2.22\,\text{mT}.$$

For comparison, a solid wire carrying the same current has $\mu_0 I/2\pi a = 2.00\,\text{mT}$ all round its surface.

18.2 The two points at 90°

Find the field at the two points on the outer surface at 90° to the line of centres.

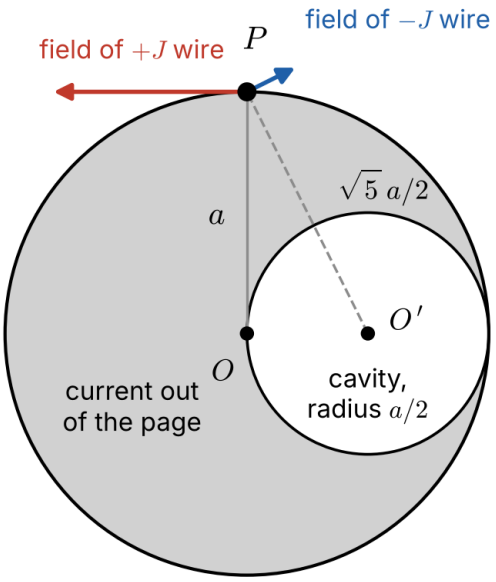


Figure 16: The point P on the outer surface at 90° to the line of centres OO', with the fields of the $+J$ and $-J$ wires there (arrow lengths proportional to field strength).

Take the wire's centre at the origin, the cavity's centre at $(a/2, 0)$, current along $+\hat{\mathbf{z}}$, and the point $P = (0, a)$.

- **Full wire** ($+J$, radius a, $\mathbf{s} = (0,a)$, $s = a$): $\hat{\mathbf{z}} \times (a\hat{\mathbf{y}}) = -a\hat{\mathbf{x}}$, so $\mathbf{B} = \dfrac{\mu_0 J a^2}{2a^2}(-a, 0) = \frac{1}{2}\mu_0 J a\,(-1, 0)$.
- **Small wire** ($-J$, radius $a/2$, $\mathbf{s} = (0,a) - (a/2, 0) = (-a/2, a)$, $s^2 = 5a^2/4$): $\hat{\mathbf{z}} \times (-\frac{a}{2}\hat{\mathbf{x}} + a\hat{\mathbf{y}}) = -\frac{a}{2}\hat{\mathbf{y}} - a\hat{\mathbf{x}} = (-a, -a/2)$, so

$$\mathbf{B} = \frac{\mu_0(-J)(a/2)^2}{2(5a^2/4)}(-a, -a/2) = -\frac{\mu_0 J}{10}(-a, -a/2) = \mu_0 J a\,\left(\tfrac{1}{10}, \tfrac{1}{20}\right).$$

Total: $\mathbf{B} = \mu_0 Ja(-0.50 + 0.10,\ 0.05) = \mu_0 Ja(-0.40,\ 0.05)$, of magnitude $\sqrt{0.16 + 0.0025}\,\mu_0 Ja = 0.403\,\mu_0 Ja$, tilted $\arctan(0.05/0.40) = 7^\circ$ from the tangent. By mirror symmetry in the line of centres, the point $(0, -a)$ has the same magnitude. In the units of Problem 18: $B_A : B_{90} : B_B = 0.25 : 0.40 : 0.42$.

18.3 Uniform transverse field and cosθ magnets

Use the uniform-field result to design a region of uniform transverse field (this is the principle of the "cosθ" dipole magnets in particle accelerators).

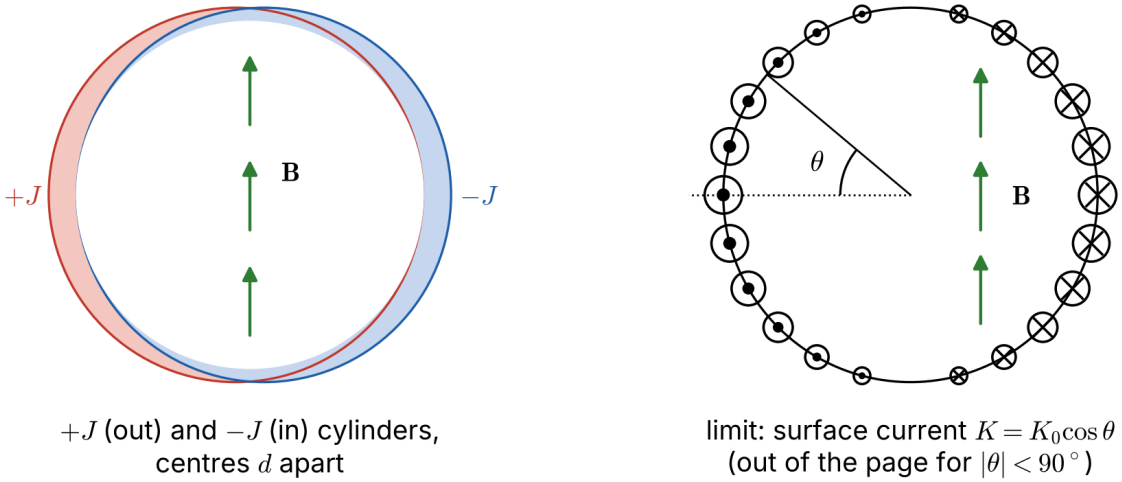


Figure 17: Left: overlapping $+J$ and $-J$ cylinders leave current only in two crescents and give a uniform **B** inside. Right: the limit $d \to 0$, a surface current $K_0 \cos\theta$ with θ measured from the side of the $+J$ crescent.

Step 1. Take two *equal* overlapping cylinders of radius R carrying $+J$ and $-J$, with centres displaced by **d**. In their overlap the currents cancel, and the remark at the end of Problem 18 gives a uniform field $\frac{1}{2}\mu_0 \mathbf{J} \times \mathbf{d}$ there. The net current flows only in two thin crescents: $+J$ where only the first cylinder is, $-J$ where only the second is.

Step 2: the limit. Let $d \to 0$ with Jd fixed. Here **d** points from the centre of the $+J$ cylinder to the centre of the $-J$ cylinder, so the $+J$ crescent lies on the $-\mathbf{d}$ side. At angle θ measured from $-\mathbf{d}$, two circles of radius R displaced by a small d are separated radially by $d\cos\theta$ (to first order in d). So the crescent has thickness $d\cos\theta$ (current out of the page for $|\theta| < 90^\circ$, into it beyond), and becomes a surface current

$$K(\theta) = J \times d\cos\theta = K_0 \cos\theta, \qquad K_0 = Jd,$$

flowing along the axis on a cylinder of radius R. Inside, the field is exactly uniform: $B = \frac{1}{2}\mu_0 Jd = \frac{1}{2}\mu_0 K_0$, perpendicular to **d**.

Application. Superconducting accelerator dipoles approximate a $\cos\theta$ current distribution with blocks of cable. For the LHC's 8.3 T, $K_0 = 2B/\mu_0 \approx 1.3 \times 10^7\,\mathrm{A\,m^{-1}}$ — an enormous current, possible only with superconductors.

Problem 19 — Pinch effect

What you need to know for these extensions

- Problem 19: pressure balance $\mathrm{d}p/\,\mathrm{d}r = -JB$ for an axial current density J and azimuthal field B.
- **Ampère's law in differential form** for an azimuthal field: $\mu_0 J = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(rB)$ (from $\oint \mathbf{B}\cdot\mathrm{d}\mathbf{l} = \mu_0 I_{\mathrm{enc}}$ applied to a thin ring).
- Integration by parts: $\int_0^a r^2 p'\,\mathrm{d}r = [r^2 p]_0^a - \int_0^a 2rp\,\mathrm{d}r$.
- **Additional knowledge for 19.1:** ideal-gas pressure $p = nk_BT$, with n counting all particles (electrons and ions).

19.1 The Bennett relation

For a plasma of temperature T and number density n, show that pressure balance gives the Bennett relation $\mu_0 I^2 = 8\pi N k_B T$, where N is the number of particles per unit length.

Step 1. Pressure balance $\mathrm{d}p/\,\mathrm{d}r = -JB$ and Ampère's law $\mu_0 J = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(rB)$ hold for *any* radial current profile. Substituting for J:

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{B}{\mu_0 r}\frac{\mathrm{d}(rB)}{\mathrm{d}r}.$$

Step 2. Multiply by r^2 and integrate over the column, $0 \le r \le a$, with $p(a) = 0$ at the edge:

$$\int_0^a r^2\frac{\mathrm{d}p}{\mathrm{d}r}\,\mathrm{d}r = -\frac{1}{\mu_0}\int_0^a rB\,\frac{\mathrm{d}(rB)}{\mathrm{d}r}\,\mathrm{d}r = -\frac{1}{2\mu_0}\left[(rB)^2\right]_0^a = -\frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi}\right)^2 = -\frac{\mu_0 I^2}{8\pi^2},$$

using $aB(a) = \mu_0 I/2\pi$ (Ampère's law round the whole column) and $\int y\,\mathrm{d}y = \frac{1}{2}y^2$ with $y = rB$. Integrating the left side by parts: $[r^2p]_0^a - \int_0^a 2rp\,\mathrm{d}r = -\int_0^a 2rp\,\mathrm{d}r$, since $p(a) = 0$. Hence

$$\int_0^a 2rp\,\mathrm{d}r = \frac{\mu_0 I^2}{8\pi^2} \quad\Longrightarrow\quad \int_0^a p\,2\pi r\,\mathrm{d}r = \frac{\mu_0 I^2}{8\pi}.$$

Step 3. For an ideal plasma $p = nk_BT$ (uniform T), and $\int_0^a n\,2\pi r\,\mathrm{d}r = N$, the number of particles per unit length:

$$Nk_BT = \frac{\mu_0 I^2}{8\pi} \quad\Longrightarrow\quad \boxed{\mu_0 I^2 = 8\pi N k_B T}$$

The current needed to confine a plasma depends only on its line density and temperature, not on the profile.

19.2 One atmosphere of pinch pressure

Find the current at which the pinch pressure at the centre of the mercury column equals atmospheric pressure.

From Problem 19, $\Delta p = \mu_0 I^2/(4\pi^2 a^2)$. Setting $\Delta p = 1.01\times10^5$ Pa and solving for I:

$$I = 2\pi a\sqrt{\frac{\Delta p}{\mu_0}} = 2\pi(5\times10^{-3})\sqrt{\frac{1.01\times10^5}{1.26\times10^{-6}}}\ \mathrm{A} = (0.0314)(2.83\times10^5)\ \mathrm{A} = 8.9\,\mathrm{kA}.$$

19.3 Hollow current shell

Repeat for a hollow cylindrical shell of current: where does the pressure change occur?

For a thin shell of radius a carrying I, Ampère's law gives $B = 0$ inside (no current enclosed) and $B = \mu_0 I/2\pi r$ outside. The force acts only on the current in the shell itself, so the pressure is uniform inside and uniform outside, with a jump *across the shell*. The surface current $K = I/2\pi a$ feels the average of the fields on its two sides, $\frac{1}{2}(0 + B_a)$ (as for the surface charge in Problem 11), so the inward force per unit area is

$$\Delta p = K \cdot \tfrac{1}{2} B_a = \frac{I}{2\pi a} \cdot \frac{\mu_0 I}{4\pi a} = \frac{B_a^2}{2\mu_0}.$$

This is the magnetic pressure in its purest form. For the mercury numbers it is 6.4 Pa, half the central excess of the solid column, but now uniform across the interior.

Problem 20 — Magnetisable spheres

What you need to know for these extensions

- Problem 20: the induced moment $\mu_0 m = 4\pi a^3(\mu - 1)B/(\mu + 2)$, the internal fields, and the dipole–dipole energy $U = \dfrac{\mu_0}{4\pi d^3}\big[\mathbf{m}_1 \cdot \mathbf{m}_2 - 3(\mathbf{m}_1 \cdot \hat{\mathbf{d}})(\mathbf{m}_2 \cdot \hat{\mathbf{d}})\big]$.
- **Bound surface current $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$** (the magnetic analogue of $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$).
- **Additional knowledge:** a uniformly magnetised sphere produces a uniform field $\frac{2}{3}\mu_0 \mathbf{M}$ inside and a pure dipole field outside (20.2); the local-field argument behind the Clausius–Mossotti relation (20.3).

20.1 Spheres side by side

Find the force when the line joining the spheres is perpendicular to $\mathbf{B}$*, and the angle at which it vanishes.*

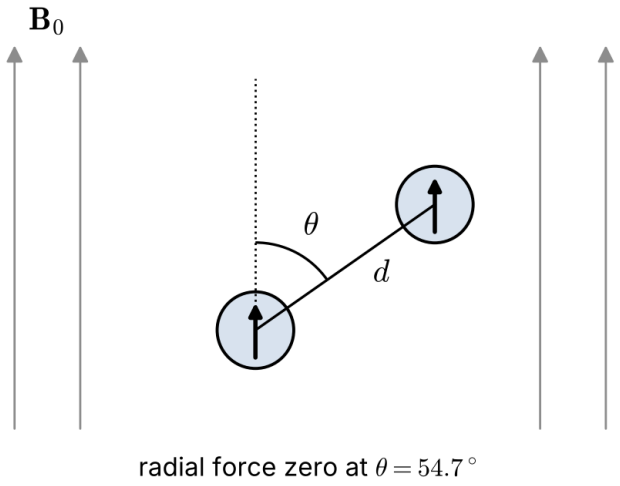


Figure 18: Two magnetised spheres a distance d apart, with the line joining them at angle θ to $\mathbf{B}_0$.

General angle. Both moments are along $\mathbf{B}$, so $\mathbf{m}_1 \cdot \mathbf{m}_2 = m^2$; if the line joining the spheres makes angle θ with $\mathbf{B}$, $\mathbf{m}_i \cdot \hat{\mathbf{d}} = m\cos\theta$. Then

$$U = \frac{\mu_0}{4\pi d^3}\left(m^2 - 3m^2\cos^2\theta\right) = \frac{\mu_0 m^2}{4\pi d^3}(1 - 3\cos^2\theta), \qquad F_r = -\frac{\partial U}{\partial d} = \frac{3\mu_0 m^2}{4\pi d^4}(1 - 3\cos^2\theta),$$

with $F_r > 0$ meaning repulsion (the force along increasing d).

- $\theta = 0$ (along $\mathbf{B}$): $F_r = \dfrac{3\mu_0 m^2}{4\pi d^4}(1-3) = -\dfrac{3\mu_0 m^2}{2\pi d^4}$, the attraction of Problem 20.
- $\theta = 90^\circ$ (side by side): $F_r = +\dfrac{3\mu_0 m^2}{4\pi d^4}$, *repulsion*, half as strong.
- $F_r = 0$ at $\cos^2\theta = \frac{1}{3}$, $\theta = 54.7^\circ$ (the "magic angle" of NMR, for the same reason). There is still a sideways force there that turns the pair towards the field direction.

20.2 The field outside, and the bound current

Find the field everywhere outside the sphere, and the bound surface current density $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ that produces it.

Outside. From Problem 20 the potential outside is the applied one plus a dipole, so the field is the applied field plus the field of a point dipole $\mathbf{m}$ at the centre:

$$\mathbf{B}_{\text{out}} = \mathbf{B}_0 + \frac{\mu_0}{4\pi}\,\frac{3(\mathbf{m}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}}{r^3}, \qquad \mu_0\mathbf{m} = \frac{4\pi a^3(\mu-1)}{\mu+2}\mathbf{B}_0.$$

Inside. $\mathbf{H}_{\text{in}} = 3\mathbf{H}_0/(\mu+2)$, so $\mathbf{M} = (\mu-1)\mathbf{H}_{\text{in}} = \dfrac{3(\mu-1)}{\mu+2}\mathbf{H}_0$, uniform, and $\mathbf{B}_{\text{in}} = \mu\mu_0\mathbf{H}_{\text{in}} = \dfrac{3\mu}{\mu+2}\mathbf{B}_0$.

Bound current. With $\mathbf{M} = M\hat{\mathbf{z}}$ and $\hat{\mathbf{n}} = \hat{\mathbf{r}}$ on the surface, $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \sin\theta\,\hat{\boldsymbol{\phi}}$ (using the Cartesian forms of $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$), so

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{r}} = M\sin\theta\,\hat{\boldsymbol{\phi}}\,:$$

an azimuthal current that varies as $\sin\theta$, like the surface current of a spinning charged sphere. This current distribution produces exactly a dipole field outside and a uniform field $\frac{2}{3}\mu_0\mathbf{M}$ inside. Check: $\mathbf{B}_0 + \frac{2}{3}\mu_0\mathbf{M} = \mathbf{B}_0\left[1 + \dfrac{2}{3}\cdot\dfrac{3(\mu-1)}{\mu+2}\right] = \mathbf{B}_0\,\dfrac{(\mu+2)+2(\mu-1)}{\mu+2} = \dfrac{3\mu}{\mu+2}\mathbf{B}_0$. ✓

20.3 The dielectric sphere and Clausius–Mossotti

Compare with a dielectric sphere in a uniform electric field: show that $(\mu-1)/(\mu+2)$ becomes $(\epsilon-1)/(\epsilon+2)$, and relate this to the Clausius–Mossotti relation.

Same mathematics. For a dielectric sphere, V plays the role of ϕ_m, and the boundary conditions (V continuous, D_r continuous) have the same form with ϵ in place of μ. So (extension 7.3)

$$\mathbf{p} = 4\pi\varepsilon_0 a^3\,\frac{\epsilon-1}{\epsilon+2}\,\mathbf{E}_0, \qquad \mathbf{P} = \frac{\mathbf{p}}{\frac{4}{3}\pi a^3} = 3\varepsilon_0\frac{\epsilon-1}{\epsilon+2}\mathbf{E}_0.$$

Clausius–Mossotti. In a dense dielectric, a molecule feels not the macroscopic field $\mathbf{E}$ but the local field $\mathbf{E}_{\text{loc}} = \mathbf{E} + \mathbf{P}/3\varepsilon_0$ — the field in a small spherical cavity, where the $\frac{1}{3}$ is the same

depolarising factor of a sphere. With N molecules per unit volume of polarisability α, $\mathbf{P} = N\alpha\mathbf{E}_{\text{loc}}$, and also $\mathbf{P} = \varepsilon_0(\epsilon - 1)\mathbf{E}$. Write $x = N\alpha/3\varepsilon_0$. Then

$$\mathbf{P} = N\alpha\left(\mathbf{E} + \frac{\mathbf{P}}{3\varepsilon_0}\right) \implies \mathbf{P}(1-x) = N\alpha\mathbf{E} \implies$$

$$\varepsilon_0(\epsilon-1)(1-x) = 3\varepsilon_0 x \implies \epsilon - 1 = x(\epsilon-1) + 3x = x(\epsilon+2),$$

$$\frac{\epsilon-1}{\epsilon+2} = \frac{N\alpha}{3\varepsilon_0}.$$

The factor $(\epsilon-1)/(\epsilon+2)$ appears in both because both come from the field of a uniformly polarised sphere.